Econ 143 — Complete Course Notes

Probability & Logic · Bayesian Inference · Bandits & Thompson Sampling
Linear Regression & OVB · Card–Krueger · Multivariate BLP
Bayesian Bootstrap · Proxy Variables · Model Selection & SURE

Lectures: 1/20/26 – 3/5/26

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Probability, Logic, and Hypothesis Testing

1 The Problem of Induction and Probabilistic Reasoning

1.1 Deductive vs. Inductive Reasoning

Classical logic operates by **deduction**: given premises that are known to be true, we derive conclusions that must be true. But in science and statistics, we reason in the opposite direction — from evidence to theory. This is **induction**, and it is fundamentally different.

Running Example.

- **Proposition A:** Newton's theory of gravity is correct.
- **Proposition B:** The centripetal acceleration of the moon is accurately predicted by Newton's theory.

We observe B is true. Does this tell us A is true? Not with certainty — but intuitively, it should increase our confidence in A. Probability is the tool we use to make this statement rigorous.

1.2 Three Forms of Argument

1.2.1 Modus Ponens (Deduction — Valid)

Modus Ponens

Premises:

(a) If A is true, then B is true $(A \Rightarrow B)$

(b) A is true

Conclusion: $\therefore$ B is true. ✓ *(logically valid)*

1.2.2 Modus Tollens / Falsification — Valid

Modus Tollens (Falsification)

Premises:

(a) If A is true, then B is true $(A \Rightarrow B)$

(b) B is false $(\neg B)$

Conclusion: $\therefore$ A is false. ✓ *(logically valid)*

This is the logic of **Popperian falsification**: a single counterexample can disprove a theory. If Newton's theory predicts the moon's acceleration and the prediction fails, we can (in principle) conclude Newton's theory is wrong.

1.2.3 Confirmation (Affirming the Consequent — Invalid but Probabilistically Useful)

Confirmation Bias / Affirming the Consequent

Premises:

(a) If A is true, then B is true $(A \Rightarrow B)$

(b) B is true

Conclusion: $\therefore$ A is true. × *(logically **invalid**!)*

However: observing B should *increase* our confidence in A. This is the role of Bayesian reasoning.

The key insight is that although "B is true $\Rightarrow$ A is true" is not deductively valid, observing evidence consistent with a theory *does* rationally increase our credence in that theory. The formal machinery to handle this is Bayes' Rule (Section 5).

2 Brief Logic Review

2.1 Propositional Logic

Let A and B be two propositions (statements that are either true or false).

Definition 2.1 (Logical Operations).

- $\neg A$ (negation): "Not A" — true when A is false.
- $A \vee B$ (disjunction): "A or B" — true when at least one of A, B is true.
- $A \wedge B$ (conjunction): "A and B" — true only when both A and B are true.

2.2 Truth Tables

Throughout, 1 denotes *true* and 0 denotes *false*.

Negation:

A	$\neg A$
1	0
0	1

Tautologies — statements that are always true regardless of the truth values of their components — are the foundation of "risk-free reasoning" (logicism, Bertrand Russell). An example is $A \vee \neg A$ (the law of excluded middle).

Conjunction ($A \wedge B$):

A	B	$A \wedge B$
1	1	1
1	0	0
0	1	0
0	0	0

Disjunction ($A \vee B$):

A	B	$A \vee B$
1	1	1
1	0	1
0	1	1
0	0	0

2.3 The Indicator Function

Logic connects cleanly to probability through the **indicator function**:

Definition 2.2 (Indicator Function)**.** For a proposition (event) A:

$$\mathbb{1}(A) = \begin{cases} 1 & \text{if } A \text{ is true} \\ 0 & \text{if } A \text{ is false (i.e., } \neg A) \end{cases}$$

More generally, for a continuous condition:

$$\mathbb{1}(X > c) = \begin{cases} 1 & \text{if } X > c \\ 0 & \text{if } X \leq c \end{cases}$$

The indicator function translates logical operations into arithmetic:

Indicator Arithmetic

$$\begin{aligned} \mathbb{1}(\neg A) &= 1 - \mathbb{1}(A) && \text{(negation)} \\ \mathbb{1}(A \wedge B) &= \mathbb{1}(A) \cdot \mathbb{1}(B) && \text{(conjunction = multiplication)} \\ \mathbb{1}(A \vee B) &= 1 - (1 - \mathbb{1}(A))(1 - \mathbb{1}(B)) && \text{(disjunction)} \\ &= \mathbb{1}(A) + \mathbb{1}(B) - \mathbb{1}(A)\mathbb{1}(B) && \text{(inclusion-exclusion)} \end{aligned}$$

The disjunction formula follows from De Morgan's law: "A or B" is the same as "not (not-A and not-B)".

3 A Brief Probability Review

3.1 The Probability Space

A **probability space** is a triple $(\Omega, \mathcal{A}, \Pr)$:

Definition 3.1 (Probability Space)**.**

1. **Sample Space** Ω: The set of all possible outcomes of an experiment (e.g., $\{H, T\}$ for a coin flip, $\mathbb{R}$ for a continuous measurement).
2. **Event Space** $\mathcal{A}$: A collection of subsets of Ω called *events*. These are the allowable events to which we can assign probabilities (technically a σ-algebra).
3. **Probability Function** $\Pr : \mathcal{A} \to \mathbb{R}$, satisfying the Kolmogorov axioms:
   (a) **Non-negativity:** For any $A \in \mathcal{A}$: $0 \leq \Pr(A) \leq 1$.
   (b) **Normality:** $\Pr(\Omega) = 1$.
   (c) **Countable Additivity:** For any countable sequence of pairwise disjoint events $A_1, A_2, \ldots \in \mathcal{A}$ (meaning $A_i \cap A_j = \emptyset$ for $i \neq j$):

$$\Pr\left(\bigcup_{i \geq 1} A_i\right) = \sum_{i \geq 1} \Pr(A_i)$$

3.2 Set Theory and Probability

The set-theoretic operations on events correspond directly to the logical operations on propositions:

- $A \cup B$ (union): the event that A *or* B occurs (at least one).
- $A \cap B$ (intersection): the event that *both* A and B occur.
- $A^c = \Omega \setminus A$ (complement): the event that A does *not* occur.

Theorem 3.2 (Complement Rule)**.**

$$\Pr(\neg A) = \Pr(A^c) = 1 - \Pr(A)$$

Proof: A and A^c are disjoint and their union is Ω, so by additivity and normality: $\Pr(A) + \Pr(A^c) = \Pr(\Omega) = 1$.

Theorem 3.3 (Inclusion-Exclusion / Addition Rule)**.**

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$$

Proof: Write $A \cup B = A \cup (B \setminus A)$ where these are disjoint. By additivity:

$$\begin{aligned}
\Pr(A \cup B) &= \Pr(A) + \Pr(B \setminus A) \\
&= \Pr(A) + \Pr(B \setminus A) + \Pr(A \cap B) - \Pr(A \cap B) \\
&= \Pr(A) + \Pr\big((B \setminus A) \cup (A \cap B)\big) - \Pr(A \cap B) \\
&= \Pr(A) + \Pr(B) - \Pr(A \cap B)
\end{aligned}$$

(using the fact that $(B \setminus A) \cup (A \cap B) = B$.)

Note: if $A \cap B = \emptyset$ (disjoint events), then $\Pr(A \cup B) = \Pr(A) + \Pr(B)$, which is just the additivity axiom.

4 Coherent Beliefs and the Dutch Book Argument

4.1 Probability as Rational Belief

The Kolmogorov axioms were introduced above as mathematical rules. But they also have a deeper normative justification: **coherent beliefs must obey the axioms of probability theory**. This is formalized by the Dutch Book argument.

Definition 4.1 (Coherence)**.** A set of beliefs is **coherent** if there is no way for a clever bookie to construct a set of bets that guarantees a profit against you regardless of the outcome.

4.2 Example of Incoherent Beliefs

Suppose someone holds these personal probability assessments:

- $\Pr(A) = 0.5$ (A: life exists on Mars)
- $\Pr(\neg A) = 0.6$

This violates the complement rule: we need $\Pr(A) + \Pr(\neg A) = 1$, but $0.5 + 0.6 = 1.1 \neq 1$. A bookie can exploit this:

- Sell you a bet: pay \$0.50 now, receive \$1 if life on Mars. Your expected gain: $1 \cdot \Pr(A) - 0.50 = 0$.
- Sell you a bet: pay \$0.60 now, receive \$1 if no life on Mars. Your expected gain: $1 \cdot \Pr(\neg A) - 0.60 = 0$.

Total cost to you: $\$0.50 + \$0.60 = \$1.10$. But exactly one of $A, \neg A$ is true, so the bookie pays out exactly \$1. The bookie profits \$0.10 with certainty. This is a **Dutch Book**.

4.3 A More Complex Dutch Book

Beliefs:

$$\begin{aligned} \Pr(A) &= 0.5 \quad \text{(life on Mars)} \\ \Pr(B) &= 0.05 \quad \text{(life on Ceres)} \\ \Pr(A \cup B) &= 0.7 \quad \text{(life on Mars or Ceres)} \end{aligned}$$

By the inclusion-exclusion formula (Theorem 3.3), coherence requires:

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B) \implies \Pr(A \cap B) = 0.5 + 0.05 - 0.7 = -0.15 < 0$$

Since probabilities cannot be negative, these beliefs are incoherent.

The Dutch Book: A bookie can offer the following three bets simultaneously:

1. Sell you: "pays \$1 if A" for \$0.50.
2. Sell you: "pays \$1 if B" for \$0.05.
3. Buy from you: "pays \$1 if $A \cup B$" for \$0.70.

Value of the Dutch Book (Bookie's profit):

$$\begin{aligned}\text{Bookie's gain} &= -0.50 + 1 \cdot \Pr(A) \\ &\quad - 0.05 + 1 \cdot \Pr(B) \\ &\quad + 0.70 - 1 \cdot \Pr(A \cup B) \\ &= 0.70 - 0.50 - 0.05 + \big[\Pr(A) + \Pr(B) - \Pr(A \cup B)\big] \\ &= 0.15 + \Pr(A \cap B) \ > \ 0\end{aligned}$$

Since $\Pr(A \cap B) \geq 0$ always, this is always positive: the bookie profits no matter what. This is called a "sure loss" bet from the perspective of the person holding the beliefs.

Dutch Book Theorem (Leonard Savage — "Probabilism")

A person's beliefs are coherent (not exploitable by a Dutch Book) **if and only if** they satisfy the Kolmogorov axioms of probability theory.

Implication: rational agents *should* have beliefs that satisfy probability theory. "Dutch Books" reveal incoherence — but note that coherence is a *minimal* standard of rationality. Coherent beliefs need not be *reasonable* beliefs (e.g., Pr(life on Mars) = 0.9999 is coherent but perhaps not well-calibrated).

5 Conditional Probability and Bayes' Rule

5.1 Conditional Probability

Definition 5.1 (Conditional Probability)**.** If $\Pr(B) > 0$, the **conditional probability** of A given B is:

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

This is the fraction of the probability mass of B that also lies in A.

Rearranging gives the **multiplication rule** (or **conjunction rule**):

$$\Pr(A \cap B) = \Pr(A \mid B) \cdot \Pr(B)$$

Logic analogy: In terms of indicator functions, conditioning is analogous to restricting attention to outcomes where B holds:

$$\mathbb{1}(A \wedge B) = \mathbb{1}(A) \cdot \mathbb{1}(B)$$

5.2 Bayes' Rule

Bayes' Rule

If $\Pr(A) > 0$, then:

$$\Pr(B \mid A) = \frac{\Pr(A \mid B) \cdot \Pr(B)}{\Pr(A)}$$

Using the **Law of Total Probability** to expand $\Pr(A)$:

$$\Pr(B \mid A) = \frac{\Pr(A \mid B) \cdot \Pr(B)}{\Pr(A \mid B) \cdot \Pr(B) + \Pr(A \mid \neg B) \cdot \Pr(\neg B)}$$

Derivation: From the multiplication rule applied both ways:

$$\Pr(A \cap B) = \Pr(A \mid B)\Pr(B) = \Pr(B \mid A)\Pr(A)$$

Solving for $\Pr(B \mid A)$ gives Bayes' Rule.

Interpretation:

- $\Pr(B)$ is the **prior** probability of B (before observing A).
- $\Pr(A \mid B)$ is the **likelihood** of observing A if B is true.
- $\Pr(B \mid A)$ is the **posterior** probability of B after observing A.
- $\Pr(A)$ is the **marginal likelihood** (normalization constant).

Bayes' Rule formalizes the confirmation logic from Section 1: observing evidence A updates our belief in hypothesis B proportionally to how much more likely A would be if B were true.

Connection to the Problem of Induction: In the Newton example:

- B = Newton's theory is correct.
- A = Moon's centripetal acceleration matches predictions.
- $\Pr(A \mid B)$ is high (if Newton's theory is correct, it predicts the moon's motion well).
- $\Pr(A \mid \neg B)$ is lower (if the theory is wrong, the prediction might still happen to match by coincidence, but less likely).
- Therefore $\Pr(B \mid A) > \Pr(B)$: observing the correct prediction raises our credence in Newton's theory.

6 Hypothesis Testing: Size, Power, and the Rare Disease Problem

6.1 The Framework

Statistical hypothesis testing is a formal procedure for deciding whether evidence warrants rejecting a null hypothesis. The two key parameters are:

Definition 6.1 (Size and Power)**.** Let H_0 be the **null hypothesis** (e.g., "the coin is fair") and H_1 be the **alternative hypothesis** (e.g., "the coin is not fair"). Let E denote the event "test result is positive (rejects H_0)", and H denote the event "H_1 is true (the effect actually exists)".

- **Size** (α): The probability of a **false positive** — incorrectly rejecting H_0 when it is actually true:
$$\alpha = \Pr(E \mid \neg H) = \Pr(\text{reject } H_0 \mid H_0 \text{ true})$$
Also called the *significance level*.

- **Power** (β): The probability of a **true positive** — correctly rejecting H_0 when H_1 is actually true:
$$\beta = \Pr(E \mid H) = \Pr(\text{reject } H_0 \mid H_1 \text{ true})$$
The *false negative rate* is $1 - \beta$ (failing to detect a real effect).

Remark 6.2. There is a fundamental trade-off: decreasing α (making it harder to reject H_0) typically also decreases β (making it less likely to detect a true effect). The choice of α and β reflects the relative costs of the two types of error.

6.2 Example: Testing a Coin for Fairness

Setup: Flip a coin 10 times. Observe all heads.

- H_0: Coin is fair ($p = 0.5$).
- There are $2^{10} = 1024$ possible sequences of heads/tails.
- Under H_0, probability of all heads: $\Pr(10 \text{ heads} \mid H_0) = (1/2)^{10} = 1/1024$.
- Probability of all heads *or* all tails under H_0: $2/1024 \approx 0.002$. Very extreme — very strong evidence against H_0.

Power calculation: Suppose the coin is actually biased with $\Pr(\text{heads}) = 0.75$. The rejection region is "all 10 heads" (or "all 10 tails"). Power:

$$\beta = \Pr(\text{all heads} \mid p = 0.75) + \Pr(\text{all tails} \mid p = 0.75) = \left(\frac{3}{4}\right)^{10} + \left(\frac{1}{4}\right)^{10} \approx 0.056$$

This is *very low power*: even if the coin is substantially biased toward heads, a test based only on the extreme event "all 10 heads" rarely detects it. In practice, rejection regions are chosen to maximize power for a given size α.

6.3 The Rare Disease Example: Bayes' Rule in Testing

This classic example illustrates why a positive test result does not mean you have the disease.

Setup:

- Disease prevalence: $\Pr(\text{disease}) = 1/1000 = 0.001$
- Test size (false positive rate): $\alpha = \Pr(\text{positive} \mid \text{no disease}) = 0.05$
- Test power (true positive rate): $\beta = \Pr(\text{positive} \mid \text{disease}) = 0.95$

Question: Given a positive test, what is the probability you actually have the disease?
By Bayes' Rule:

$$\Pr(\text{disease} \mid \text{positive}) = \frac{\Pr(\text{positive} \mid \text{disease}) \cdot \Pr(\text{disease})}{\Pr(\text{positive})}$$

First, compute Pr(positive) via the Law of Total Probability:

$$\begin{aligned}\Pr(\text{positive}) &= \Pr(\text{positive} \mid \text{disease})\Pr(\text{disease}) + \Pr(\text{positive} \mid \text{no disease})\Pr(\text{no disease}) \\ &= 0.95 \times 0.001 + 0.05 \times 0.999 \\ &= 0.00095 + 0.04995 = 0.05090\end{aligned}$$

Therefore:

$$\Pr(\text{disease} \mid \text{positive}) = \frac{0.95 \times 0.001}{0.05090} \approx \frac{0.00095}{0.05090} \approx 0.019$$

Key Takeaway: Base Rate Neglect

Even with a high-power ($\beta = 0.95$), low-size ($\alpha = 0.05$) test, a positive result for a *rare* disease only has a $\approx 2\%$ posterior probability of being a true positive. The vast majority of positive results are false positives because the disease is so rare. This is the **base rate fallacy**: ignoring Pr(disease) when interpreting test results.

6.4 Size of the Rare Disease Test (Using Bayes)

The quantity "size" in this context can be computed as:

$$\Pr(\text{disease} \mid \text{positive}) = \frac{\beta \cdot \frac{1}{1000}}{\beta \cdot \frac{1}{1000} + \alpha \cdot \frac{999}{1000}} \approx 0.02$$

This is the probability that a detected disease is real, given the prevalence and the test characteristics.

7 Program Evaluation and the Bayesian Interpretation of p-values

7.1 Motivating Problem

Consider the evaluation of randomized controlled trials (RCTs) in a domain (e.g., policy programs). The statistical design uses:

- $\alpha = 0.05$ (significance level / size)
- $\beta = 0.80$ (power — probability of detecting a true effect)

Empirically: about 1/2 of RCTs in this domain reject the null of no effect (i.e., about half report statistically significant results).

Question: What fraction p of these programs actually have a real effect?

Setup:

- $\Pr(\text{reject } H_0 \mid \text{real effect}) = \beta = 0.80$
- $\Pr(\text{reject } H_0 \mid \text{no real effect}) = \alpha = 0.05$
- Let p = prior probability a program has a real effect.

The fraction of studies that reject H_0 is:

$$\Pr(\text{reject}) = \beta \cdot p + \alpha \cdot (1-p) = 0.80p + 0.05(1-p)$$

Setting this equal to the observed 1/2:

$$\frac{1}{2} = 0.80p + 0.05(1-p) \implies \frac{1}{2} = 0.75p + 0.05 \implies p = \frac{0.45}{0.75} = \frac{3}{5} = 0.60$$

So about 60% of the programs in this domain have a genuine effect.

7.2 False Discovery Rate

A crucial quantity for interpreting scientific results is: given that a study *did not* reject the null, what is the probability the program truly has no effect?

$$\Pr(\text{no actual effect} \mid \text{no effect found}) = \frac{\Pr(\text{no effect found} \mid \text{no actual effect}) \cdot \Pr(\text{no actual effect})}{\Pr(\text{no effect found})}$$

$$\begin{aligned}
\Pr(\text{no effect found} \mid \text{no actual effect}) &= 1 - \alpha = 0.95 \\
\Pr(\text{no effect found} \mid \text{actual effect}) &= 1 - \beta = 0.20 \\
\Pr(\text{no actual effect}) &= 1 - p = 0.40 \\
\Pr(\text{no effect found}) &= (1-\beta)p + (1-\alpha)(1-p) = 0.20 \times 0.60 + 0.95 \times 0.40 = 0.50
\end{aligned}$$

$$\Pr(\text{no actual effect} \mid \text{no effect found}) = \frac{0.95 \times 0.40}{0.50} = 0.76$$

Equivalently, the false negative rate (probability a genuine effect was missed) is $1-0.76 = 0.24$.

Sensitivity to Prior p

- With $p = 0.60$ (computed above): $\Pr(\text{no actual effect} \mid \text{no effect found}) \approx 0.76$
- With $p = 0.90$ (optimistic prior): $\Pr(\text{no actual effect} \mid \text{no effect found}) \approx 0.35$
- With $\beta = 0.95$, $\alpha = 0.01$, $p = 3/5$: $\Pr(\text{no actual effect} \mid \text{no effect found}) \approx 0.91$

The posterior probability that a non-rejected null is actually null depends heavily on the prior probability of effectiveness p. Higher-powered tests (large β) and lower significance levels (small α) reduce the false negative rate substantially.

7.3 General Formula

The general expression for $\Pr(\text{no actual effect} \mid \text{no effect found})$:

$$\Pr(\text{no actual effect} \mid \text{no effect found}) = \frac{(1-\alpha)(1-p)}{(1-\beta)p + (1-\alpha)(1-p)}$$

This highlights the three key drivers:

1. **Prior** p: How often programs in this domain really work.
2. **Size** α: How often we false-alarm. Smaller $\alpha \Rightarrow$ fewer false positives.
3. **Power** β: How reliably we detect real effects. Larger $\beta \Rightarrow$ fewer false negatives.

8 Summary Table

Concept	Formula / Statement	Interpretation
Negation	$\Pr(\neg A) = 1 - \Pr(A)$	Complement rule
Addition	$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$	Inclusion-exclusion
Conjunction	$\Pr(A \cap B) = \Pr(A \mid B)\Pr(B)$	Multiplication rule
Bayes' Rule	$\Pr(B \mid A) = \frac{\Pr(A \mid B)\Pr(B)}{\Pr(A)}$	Posterior from prior + likelihood
Size α	$\alpha = \Pr(E \mid \neg H)$	False positive rate
Power β	$\beta = \Pr(E \mid H)$	True positive rate
Dutch Book	Beliefs must satisfy Kolmogorov axioms	Coherence requirement
False Discovery	$\Pr(\neg H \mid \neg E) = \frac{(1-\alpha)(1-p)}{(1-\beta)p + (1-\alpha)(1-p)}$	Prob. non-rejection is correct

Big Picture

Probability serves two roles: (1) as a mathematical model for random experiments (frequentist view), and (2) as a representation of rational degrees of belief (Bayesian / subjectivist view). The Dutch Book argument shows these are not separate: any agent whose beliefs violate the probability axioms can be exploited by a clever adversary. Bayes' Rule then provides the correct procedure for updating beliefs in light of new evidence — formalizing the informal intuition that seeing B should increase your confidence in A when A predicts B.

Bayesian Prediction, Bernoulli Bandits, and Conjugate Priors

9 A Bayesian Prediction Problem: The Lindy Effect

9.1 Setup

Consider the following prediction problem. You meet a random person on the street. You observe that they are currently age t. What is your best prediction of their total lifespan t_{total}?

Definition 9.1 (Variables)**.**

- t_{total}: the person's total life expectancy (unknown to you).
- t: their current age at the time you encounter them (observed).

This is a genuine prediction problem: you observe t and want to form a posterior belief about t_{total}.

9.2 The Likelihood: Uniform Distribution Conditional on Lifespan

Given that someone lives to age t_{total}, at what age are you equally likely to encounter them? If meetings are uniformly distributed over their lifetime, the **likelihood** of observing them at age t given their lifespan is t_{total} is:

$$p(t \mid t_{\text{total}}) = \begin{cases} \dfrac{1}{t_{\text{total}}} & \text{if } t_{\text{total}} \geq t \\ 0 & \text{otherwise} \end{cases}$$

This is a **uniform likelihood**: conditional on living to t_{total}, each age $t \leq t_{\text{total}}$ is equally probable as the observed age.

9.3 The Prior: Pareto Distribution

What do we believe about t_{total} before observing t? A natural prior for quantities like lifespans (which have heavy tails) is the **Pareto distribution**:

Definition 9.2 (Pareto Distribution)**.** A random variable T has a Pareto distribution with shape parameter $\gamma > 0$ if:

$$p(t_{\text{total}}) \propto t_{\text{total}}^{-\gamma}, \quad t_{\text{total}} \geq t_{\min}$$

The Pareto distribution is a **power law**: it assigns relatively high probability to large values, making it appropriate for heavy-tailed quantities like city populations, wealth, or life spans of ideas, companies, and people.

Why the Pareto prior?

- It is scale-free (no characteristic scale), consistent with the observation that many natural and social phenomena follow power laws.
- It will yield an analytically tractable posterior when combined with the uniform likelihood (see below).

- The **Lindy Effect** (named after the Lindy's deli in New York) is the empirical observation that for non-perishable things (books, technologies, ideas), future life expectancy increases with current age. The Pareto prior combined with a uniform likelihood generates exactly this effect.

9.4 Bayes' Rule: Deriving the Posterior

By Bayes' Rule, the **posterior density** of t_{total} given that we observe the person at age t is:

$$p(t_{\text{total}} \mid t) = \frac{p(t \mid t_{\text{total}}) \cdot p(t_{\text{total}})}{p(t)}$$

where the denominator is the marginal likelihood (normalizing constant):

$$p(t) = \int p(t, t_{\text{total}})\, dt_{\text{total}} = \int p(t \mid t_{\text{total}}) \cdot p(t_{\text{total}})\, dt_{\text{total}}$$

Computing the normalizing constant:

$$\begin{aligned} p(t) &= \int_t^\infty \frac{1}{t_{\text{total}}} \cdot t_{\text{total}}^{-\gamma}\, dt_{\text{total}} = \int_t^\infty t_{\text{total}}^{-(1+\gamma)}\, dt_{\text{total}} \\ &= \left[-\frac{1}{\gamma} t_{\text{total}}^{-\gamma} \right]_t^\infty = \frac{1}{\gamma} t^{-\gamma} \end{aligned}$$

Computing the posterior density:

$$p(t_{\text{total}} \mid t) = \frac{t_{\text{total}}^{-(1+\gamma)}}{\frac{1}{\gamma} t^{-\gamma}} = \frac{\gamma\, t^\gamma}{t_{\text{total}}^{\gamma+1}}$$

Posterior Density (Pareto Posterior)

$$p(t_{\text{total}} \mid t) = \frac{\gamma\, t^\gamma}{t_{\text{total}}^{\gamma+1}}, \qquad t_{\text{total}} \geq t$$

This is itself a **Pareto distribution** with shape parameter γ and minimum t. The prior-to-posterior update preserves the Pareto family — the Pareto distribution is the *conjugate prior* for the uniform likelihood on $[0, t_{\text{total}}]$.

9.5 Posterior CDF and the Lindy Effect

The posterior probability that t_{total} exceeds some threshold t^* (given observed age t) is:

$$\Pr(t_{\text{total}} > t^* \mid t) = \int_{t^*}^\infty p(t_{\text{total}} \mid t)\, dt_{\text{total}} = \int_{t^*}^\infty \frac{\gamma\, t^\gamma}{t_{\text{total}}^{\gamma+1}}\, dt_{\text{total}} = \left(\frac{t}{t^*} \right)^\gamma$$

The Posterior Median. Set $\Pr(t_{\text{total}} > t^* \mid t) = \frac{1}{2}$ to find the posterior median belief about total lifespan:

$$\left(\frac{t}{t^*}\right)^\gamma = \frac{1}{2} \implies t^* = 2^{1/\gamma} \cdot t$$

The Lindy Effect

The posterior median lifespan given observed age t is:

$$t^* = 2^{1/\gamma} \cdot t$$

Since $2^{1/\gamma} > 1$ for any $\gamma > 0$, the predicted total lifespan is **proportional to current age**. A person you encounter at age 50 has a higher predicted total lifespan than one you encounter at age 20. The older something already is, the longer it is expected to continue to live/exist. This is the Lindy Effect.

Note: The posterior CDF $\Pr(t_{\text{total}} > t^* \mid t) = (t/t^*)^\gamma$ is analogous to a **survival function**, giving the probability of "surviving" beyond a given future age. The Pareto distribution is the continuous analogue of the geometric distribution (memoryless property for discrete distributions).

10 Bernoulli Bandits: Setup

10.1 The Multi-Armed Bandit Framework

The **multi-armed bandit** (MAB) is a sequential decision-making problem. The name comes from a casino metaphor: a gambler faces a row of slot machines ("one-armed bandits"), each with an unknown payout probability, and must decide which to play at each round to maximize total reward.

Definition 10.1 (Bernoulli Bandit)**.** There are $J + 1$ **arms** (or treatments), indexed $j = 0, 1, \ldots, J$. Each arm has an unknown success probability $\alpha_j \in [0, 1]$. At each round $t = 1, \ldots, T$, the agent pulls one arm and observes a binary reward:

$$Y_j(t) \sim \text{Bernoulli}(\alpha_j)$$

The **best arm** is $j^* = \arg\max_j \alpha_j$ (the one with the highest success probability α_{j^*}).

Motivating Application: A/B Testing on the Web.

- $Y_0, Y_1, \ldots, Y_J$ are different web page designs shown to customers.
- $Y_j(t) = 1$ if customer clicks (success), $Y_j(t) = 0$ if they do not (failure).
- $\alpha_j = \Pr(\text{click} \mid \text{shown page } j)$.
- The goal is to identify and serve the page with the highest click-through rate while minimizing total losses along the way.

Multi-armed bandits are also used in clinical trials (identifying the best treatment), pricing experiments, and any sequential decision environment with uncertain payoffs.

10.2 Decision Variable and History

At each round t, the agent selects exactly one arm. Define:

$$D_j(t) = \mathbb{1}[\text{arm } j \text{ is chosen in round } t], \qquad D(t) = (D_0(t), D_1(t), \ldots, D_J(t))$$

so $\sum_j D_j(t) = 1$ (only one arm is pulled per round).

The agent maintains running statistics for each arm:

$$S_j(t) = \sum_{s=1}^{t-1} D_j(s) \cdot Y_j(s) \qquad (\text{\# of successes for arm } j \text{ through round } t-1)$$

$$F_j(t) = \sum_{s=1}^{t-1} D_j(s) \cdot (1 - Y_j(s)) \qquad (\text{\# of failures for arm } j \text{ through round } t-1)$$

$$N_j(t) = S_j(t) + F_j(t) = \sum_{s=1}^{t-1} D_j(s) \qquad (\text{total pulls of arm } j \text{ through round } t-1)$$

Casino analogy: Each arm is like a slot machine. Each time you pull arm j ("put a coin in"), you either win ($Y_j = 1$) or lose ($Y_j = 0$). S_j and F_j track your wins and losses on machine j.

10.3 Researcher Prior Beliefs

Because the α_j are unknown, the researcher places a **Beta prior** over each one (independently):

$$\alpha_j \sim \text{Beta}(a_j, b_j), \qquad j = 0, 1, \ldots, J$$

The standard uninformative choice is $a_j = b_j = 1$ (uniform prior, Beta(1, 1)), which assigns equal probability to all values $\alpha_j \in [0, 1]$.

After observing the history $\mathcal{I}_{t-1}$ (all arms pulled and their outcomes through round $t-1$), the **posterior** by Beta–Bernoulli conjugacy is:

$$\alpha_j \mid \mathcal{I}_{t-1} \sim \text{Beta}(S_j(t) + a_j, \;\; F_j(t) + b_j)$$

The prior parameters (a_j, b_j) act as "pseudo-counts" of prior successes and failures.

10.4 Thompson Sampling (Thompson, 1935)

How should the agent select arms at each round? The **Thompson Sampling** algorithm answers this:

Thompson Sampling

At each round $t = 1, \ldots, T$:

1. For each arm $j = 0, \ldots, J$, draw a sample from the beginning-of-round-t posterior:
$$\tilde{\alpha}_j(t) \sim \text{Beta}(S_j(t) + a_j,\ F_j(t) + b_j)$$
2. Select the arm with the highest posterior draw:
$$j^*(t) = \arg\max_{j \in \{0,\ldots,J\}} \tilde{\alpha}_j(t)$$
3. Pull arm $j^*(t)$, observe $Y_{j^*(t)}(t)$, and update S_{j^*} or F_{j^*}.

Intuition with 3 arms ($j = 0, 1, 2$): After $t = 361$ rounds, the three posterior distributions $\text{Beta}(S_j + a_j, F_j + b_j)$ are concentrated in different regions of $[0, 1]$. In each new round, one sample is drawn from each — arm $j = 1$ might have a tight posterior around 0.7, arm $j = 2$ around 0.5, and arm $j = 0$ a broad, flat posterior (less data). Whichever has the highest realization in a given round gets played.

Key property: Thompson Sampling naturally balances exploration and exploitation. Arms with high posterior mean get played often (exploitation); arms with high posterior uncertainty occasionally produce a large draw, prompting exploration. As more data is gathered, the posterior for the best arm concentrates, and it gets played almost exclusively.

11 Bayesian vs. Frequentist Inference

11.1 The Fundamental Distinction

Both paradigms concern inference about a parameter θ (e.g., the success probability of an arm, or a treatment effect).

Definition 11.1 (Frequentist vs. Bayesian).

- **Frequentist**: θ is a **fixed but unknown constant**. Probability refers only to the long-run frequency of repeated experiments. We cannot assign a probability distribution to θ itself.
- **Bayesian**: θ is a **random variable** with a **prior distribution** $\pi(\theta)$ reflecting beliefs before seeing data. After observing data x, beliefs are updated to the **posterior** $\pi(\theta \mid x)$.

Both frameworks assume data are drawn i.i.d. from $f(x \mid \theta)$.

11.2 Bayesian Updating via Bayes' Rule

Given data $x = (x_1, \ldots, x_n)$ drawn i.i.d. from $f(x \mid \theta)$ and prior $\pi(\theta)$:

$$\pi(\theta \mid x) = \frac{f(x \mid \theta) \cdot \pi(\theta)}{f(x)} = \frac{f(x \mid \theta) \cdot \pi(\theta)}{\int f(x \mid \theta')\, \pi(\theta')\, d\theta'}$$

Two approaches to point estimation from the posterior:

- **Bayesian (posterior mean):** $\hat{\theta}_{\text{Bayes}} = \mathbb{E}[\theta \mid x]$ — minimizes posterior expected squared error.
- **Maximum Likelihood (MLE):** $\hat{\theta}_{\text{MLE}} = \arg\max_\theta f(x \mid \theta)$ — maximizes the likelihood of the observed data. No prior is used; this is the frequentist estimator.

The Core Distinction

- **Bayesian:** data changes your *probability distribution* over θ. After seeing data, you have a posterior $\pi(\theta \mid x)$, not just a point estimate.
- **Frequentist:** data gives you a point estimate $\hat{\theta}$ with a sampling distribution. No probability is assigned to θ directly.

12 Conjugate Priors: The Beta–Binomial Family

12.1 Why Conjugate Priors?

Computing the posterior $\pi(\theta \mid x)$ generally requires evaluating the integral $\int f(x \mid \theta)\pi(\theta)\, d\theta$, which is analytically intractable in most cases. A **conjugate prior** is chosen so that the posterior belongs to the same distributional family as the prior, making the update closed-form.

Definition 12.1 (Conjugate Prior)**.** A prior $\pi(\theta)$ is **conjugate** to a likelihood $f(x \mid \theta)$ if the posterior $\pi(\theta \mid x) \propto f(x \mid \theta)\, \pi(\theta)$ belongs to the same parametric family as $\pi(\theta)$.

12.2 The Beta–Binomial Conjugate Pair

Setup:

- **Likelihood:** Each observation $x_i \in \{0, 1\}$ is Bernoulli(θ). With n observations, the sum $\sum x_i \sim \text{Binomial}(n, \theta)$:

$$f(x_i \mid \theta) = \theta^{x_i}(1-\theta)^{1-x_i}$$

- **Prior:** $\theta \sim \text{Beta}(\alpha, \beta)$ with pdf proportional to $\theta^{\alpha-1}(1-\theta)^{\beta-1}$.

Posterior derivation: The joint likelihood of n i.i.d. observations is:

$$f(x_1, \ldots, x_n \mid \theta) = \prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i} = \theta^{\sum x_i}(1-\theta)^{n-\sum x_i}$$

The unnormalized posterior is:

$$\begin{aligned}\pi(\theta \mid x_1, \ldots, x_n) &\propto f(x_1, \ldots, x_n \mid \theta) \cdot \pi(\theta) \\ &\propto \theta^{\sum x_i}(1-\theta)^{n-\sum x_i} \cdot \theta^{\alpha-1}(1-\theta)^{\beta-1} \\ &= \theta^{\sum x_i+\alpha-1}(1-\theta)^{n-\sum x_i+\beta-1}\end{aligned}$$

Recognizing this as the kernel of a Beta distribution:

Beta–Binomial Conjugacy

$$\theta \mid x_1, \ldots, x_n \sim \text{Beta}\left(\sum_{i=1}^{n} x_i + \alpha,\ \ n - \sum_{i=1}^{n} x_i + \beta\right)$$

The posterior mean is:

$$\mathbb{E}[\theta \mid x] = \frac{\sum x_i + \alpha}{n + \alpha + \beta}$$

Interpretation: The posterior parameters are simply the prior pseudo-counts (α, β) plus the observed counts of successes and failures $(\sum x_i,\ n - \sum x_i)$. No numerical integration is needed.

Posterior mean intuition:

$$\mathbb{E}[\theta \mid x] = \frac{\text{prior successes} + \text{observed successes}}{\text{prior total} + \text{observed total}} \approx \frac{\sum x_i}{n} \quad \text{for large } n$$

As $n \to \infty$, the posterior mean converges to the sample proportion (the MLE), and the prior becomes negligible. With little data, the prior pulls the estimate toward $\alpha/(\alpha+\beta)$.

Connection to the Bandit: In the Bernoulli bandit, arm j starts with prior $\text{Beta}(a_j, b_j)$. After S_j successes and F_j failures, the posterior is $\text{Beta}(S_j + a_j, F_j + b_j)$, which is exactly the Beta–Binomial conjugate update with $\sum x_i = S_j$, $n = N_j$, $\alpha = a_j$, $\beta = b_j$.

13 Bayesian Rulers and Sure Losers

13.1 Setup: Hypothesis Testing as a Betting Game

This section connects Bayesian coherence (Dutch Books) with hypothesis testing. Consider 1000 trials of an experiment. Define:

- $H = 1$: the null hypothesis is **false** (effect exists); $H = 0$: null is true.

- $E = 1$: the test **rejects** the null; $E = 0$: fails to reject.
- $\alpha = P(E = 1 \mid H = 0) = P(E \mid \neg H)$: **size** (false positive rate). We want α small.
- $\beta = P(E = 1 \mid H = 1) = P(E \mid H)$: **power** (true positive rate). We want β large.
- $\gamma = P(H = 1)$: **prior probability** that the effect exists.

13.2 Posterior Probabilities via Bayes

(a) Probability the effect exists given the test rejects ($E = 1$):

$$P(H \mid E) = \frac{P(E \mid H)\, P(H)}{P(E \mid H)\, P(H) + P(E \mid \neg H)\, P(\neg H)} = \frac{\beta\gamma}{\beta\gamma + \alpha(1-\gamma)}$$

(b) Probability the effect exists given the test fails to reject ($E = 0$):

$$P(H \mid \neg E) = \frac{P(\neg E \mid H)\, P(H)}{P(\neg E \mid H)\, P(H) + P(\neg E \mid \neg H)\, P(\neg H)} = \frac{(1-\beta)\gamma}{(1-\beta)\gamma + (1-\alpha)(1-\gamma)}$$

These two quantities — the *positive predictive value* and the *false omission rate* — are what Bayesian analysis provides beyond the frequentist p-value. They depend critically on the prior γ.

13.3 The Sure-Loser (Dutch Book) Analysis

Now suppose a bookie offers a betting scheme based on these tests. Over 1000 trials, define three types of false positive bets:

False Positive calculations (out of 1000 trials):

- FP_1: Trials where $H = 1$ (effect exists) but $E = 0$ (missed):

$$FP_1 = 1000 \cdot P(H \cap \neg E) = 1000 \cdot P(H \mid \neg E) \cdot P(\neg E)$$

- FP_2: Trials where $H = 1$ (effect exists) and $E = 1$ (correctly detected):

$$FP_2 = 1000 \cdot P(H \mid E) \cdot P(\neg E)$$

- FP_3: Trials where the test rejects ($E = 1$):

$$FP_3 = 1000\varepsilon \cdot P(E)$$

Total false positive count:

$$\begin{aligned} FP &= FP_1 + FP_2 + FP_3 \\ &= 1000 \cdot P(H \mid E)\,[P(E) + P(\neg E)] \\ &= 1000\varepsilon \cdot P(E) \\ &= \frac{1000\beta\gamma}{\beta\gamma + \alpha(1-\gamma)} + 1000\varepsilon \cdot [\beta\gamma + \alpha(1-\gamma)] \end{aligned}$$

(c) Total winnings when the null is not rejected ($E = 0$): The total winnings in the non-rejection case are:

$$\begin{aligned}
\text{Total winnings} &= 1000 \cdot P(H \mid E) - (FP_1 + FP_2 + FP_3) \\
&= \frac{1000\beta\gamma}{\beta\gamma + \alpha(1-\gamma)} - \left(\frac{1000\beta\gamma}{\beta\gamma + \alpha(1-\gamma)} + 1000\varepsilon \cdot [\beta\gamma + \alpha(1-\gamma)] \right) \\
&= -1000\varepsilon \cdot [\beta\gamma + \alpha(1-\gamma)] \\
&= (\beta - \alpha)\gamma + \alpha
\end{aligned}$$

(d) When is there a sure loss? The expression $(\beta - \alpha)\gamma + \alpha$ is always positive when $\beta > \alpha$ (power exceeds size). This means: if you use a test with more power than size, but your prior γ is not coherently aligned with the test's properties, a bookie can construct bets that guarantee a profit from you.

Sure-Loser Theorem

Main idea: Incoherent beliefs — specifically, using a hypothesis test whose implied probability assessments violate Bayes' Rule — can be mathematically exploited. A Dutch Book (sure-loss bet) exists whenever your stated prior γ and your acting α, β are inconsistent.

Implication: We should hold **coherent beliefs** aligned with the laws of probability. Using a test with size α and power β implicitly commits you to specific posterior beliefs via Bayes' Rule. Treating the p-value as a direct probability that the null is true (without accounting for the prior γ) is precisely the kind of incoherence that can be exploited.

Practical consequence: The sure-loser analysis formalizes the message from the program evaluation example in the previous lecture notes: whether or not a statistically significant result implies a real effect depends critically on the prior probability γ that effects exist in your domain. Ignoring γ is not just philosophically wrong — it is exploitable.

14 Summary Table

Object	Formula	Interpretation
Posterior (Pareto)	$p(t_{\text{total}} \mid t) = \dfrac{\gamma\, t^{\gamma}}{t_{\text{total}}^{\gamma+1}}$	Pareto posterior; conjugate to uniform likelihood
Lindy Median	$t^* = 2^{1/\gamma} \cdot t$	Predicted lifespan $\propto$ observed age
Bandit posterior	$\alpha_j \mid \mathcal{I}_{t-1} \sim \text{Beta}(S_j + a_j, F_j + b_j)$	Beta–Bernoulli update
Thompson Sampling	$j^*(t) = \arg\max_j \tilde{\alpha}_j(t)$	Choose arm with highest posterior draw
Beta–Binomial	$\theta \mid x \sim \text{Beta}(\sum x_i + \alpha,\ n - \sum x_i + \beta)$	Conjugate update
Posterior mean	$\mathbb{E}[\theta \mid x] = \dfrac{\sum x_i + \alpha}{n + \alpha + \beta}$	Shrinks MLE toward prior
PPV	$P(H \mid E) = \dfrac{\beta\gamma}{\beta\gamma + \alpha(1-\gamma)}$	Posterior prob. of effect given rejection
Sure-loser	$(\beta - \alpha)\gamma + \alpha > 0$	Incoherent priors are exploitable

Big Picture

These notes connect three threads: (1) Bayesian prediction using Bayes' Rule to update a prior (Pareto) with a likelihood (uniform), yielding the Lindy Effect as a natural consequence. (2) Multi-armed bandits as a repeated Bayesian learning problem, where Thompson Sampling draws from posteriors to balance exploration and exploitation. (3) Conjugate priors (Beta–Binomial) that make posterior computation tractable in closed form. The sure-loser theorem ties everything back to the Dutch Book / coherence argument: using hypothesis tests without accounting for prior probabilities leads to incoherent beliefs that can be exploited by a rational adversary.

Thompson Sampling, Regret Bounds, and MCMC Cipher Decoding

15 Multi-Armed Bandits: Setup and Framework

The **multi-armed bandit problem** is one of the canonical problems in sequential decision-making and online learning. The name comes from the image of a gambler at a row of slot machines ("one-armed bandits"), each with an unknown payout probability, deciding which arms to pull in sequence to maximize total reward.

15.1 The Basic Setup

We have $J+1$ arms, indexed $j = 0, 1, \ldots, J$. Each arm j has an unknown Bernoulli success probability $\alpha_j \in [0, 1]$:

$$Y_j(t) \sim \text{Bernoulli}(\alpha_j), \qquad j = 0, 1, \ldots, J$$

where $Y_j(t) \in \{0, 1\}$ is the reward (success or failure) from pulling arm j at time t, and the α_j are the **unknown success probabilities** (the "likelihood" for each arm).

Key Idea: Exploration vs. Exploitation

At each round, the agent must decide:

- **Exploit:** Pull the arm that currently looks best (maximize immediate reward).
- **Explore:** Pull a less-certain arm to learn more about it (potentially improve future decisions).

A purely greedy (exploit-only) algorithm does not explore enough and gets stuck on a suboptimal arm. Good algorithms balance both.

15.2 Bayesian Prior: The Beta Distribution

Because each α_j is unknown, we place a Bayesian prior over it. The **Beta distribution** is the natural conjugate prior for a Bernoulli likelihood:

$$\alpha_j \sim \text{Beta}(a_j,\, b_j), \qquad j = 0, 1, \ldots, J$$

where the priors are assumed **independent** across arms. The parameters $a_j > 0$ and $b_j > 0$ encode our prior beliefs about arm j.

Definition 15.1 (Beta Distribution)**.** A random variable $X \sim \text{Beta}(a, b)$ has pdf

$$f(x; a, b) = \frac{x^{a-1}(1-x)^{b-1}}{B(a, b)}, \quad x \in [0, 1]$$

where $B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$ is the Beta function. Its mean and variance are:

$$\mathbb{E}[X] = \frac{a}{a+b}, \qquad \mathbb{V}[X] = \frac{ab}{(a+b)^2(a+b+1)}.$$

The typical uninformative (uniform) prior is $\text{Beta}(1, 1)$, which is a uniform distribution on $[0, 1]$.

16 Bayesian Updating in the Bandit

16.1 History Statistics

At time t, the agent's **information set** $\mathcal{I}_{t-1}$ contains the history of all past decisions and outcomes. For each arm j, we track two sufficient statistics:

$$S_j(t) = \sum_{s=1}^{t-1} D_j(s) \cdot Y_j(s) \qquad (\text{\# of previous successes for arm } j)$$

$$F_j(t) = \sum_{s=1}^{t-1} D_j(s) \cdot \big(1 - Y_j(s)\big) \qquad (\text{\# of previous failures for arm } j)$$

where $D_j(s) = \mathbb{1}[\text{arm } j \text{ was pulled at time } s]$.
The total number of pulls for arm j up to time t is:

$$N_j(t) = S_j(t) + F_j(t) = \sum_{s=1}^{t-1} D_j(s).$$

16.2 Posterior Distribution

Because the Beta distribution is the conjugate prior for the Bernoulli likelihood, the posterior after observing the history is simply:

Bayesian Updating Rule (Beta-Bernoulli Conjugacy)

$$\alpha_j \mid \mathcal{I}_{t-1} \;\sim\; \text{Beta}\big(S_j(t) + a_j,\;\; F_j(t) + b_j\big)$$

This is the agent's **posterior belief** about arm j's success probability at the beginning of round t. The initial prior parameters a_j, b_j act as "pseudo-counts" of prior successes and failures.

16.3 Posterior Mean and Variance

From the Beta distribution formulas, the posterior mean and variance for arm j at time t are:

$$\mathbb{E}[\alpha_j \mid \mathcal{I}_{t-1}] = \frac{S_j(t) + a_j}{N_j(t) + a_j + b_j} \approx \frac{\text{number of successes}}{\text{total number of attempts}}$$

$$\mathbb{V}[\alpha_j \mid \mathcal{I}_{t-1}] = \frac{(S_j(t) + a_j)(F_j(t) + b_j)}{(N_j(t) + a_j + b_j)^2 \cdot (N_j(t) + a_j + b_j + 1)}$$

Remark 16.1. As $N_j(t) \to \infty$ (i.e., the arm is pulled more), the posterior variance $\to 0$. The more we try an arm, the lower our uncertainty becomes. Conversely, arms that have been pulled very few times have high posterior variance, reflecting genuine uncertainty.

17 Thompson Sampling

17.1 The Algorithm

Thompson Sampling (Posterior Sampling)

At each round $t = 1, \ldots, T$:

1. For each arm $j = 0, \ldots, J$, draw a sample from the current posterior:

$$\hat{\alpha}_j \sim \text{Beta}\big(S_j(t) + a_j,\ F_j(t) + b_j\big)$$

2. Select the arm with the highest posterior draw:

$$j^*(t) = \arg\max_{j \in \{0,\ldots,J\}} \hat{\alpha}_j$$

3. Pull arm $j^*(t)$, observe $Y_{j^*(t)}(t) \in \{0, 1\}$, and update S_{j^*} or F_{j^*} accordingly.

17.2 Why Thompson Sampling Works

Thompson Sampling achieves a natural exploration-exploitation balance:

- **Arms with high posterior mean** are frequently drawn with large $\hat{\alpha}_j$, so they tend to be selected often (exploitation).
- **Arms with high posterior variance** (i.e., under-explored arms) occasionally produce a large draw, causing them to be selected for exploration.
- As an arm is pulled more, its posterior variance shrinks, reducing the chance it gets a lucky draw unless it truly has a high mean.

17.3 Comparison: Greedy Algorithm

The **greedy algorithm** simply picks $j^*(t) = \arg\max_j \mathbb{E}[\alpha_j \mid \mathcal{I}_{t-1}]$ at each round—always choosing the arm with the highest posterior mean.

- **Problem:** It does not explore. Once an arm appears best based on limited data, the algorithm locks onto it and never discovers whether other arms might actually be better.
- **Result:** The greedy algorithm can suffer linear regret (i.e., it never converges to the optimal arm if the initial data misled it).

Thompson Sampling, by contrast, is guaranteed to have *sublinear* regret (see Section 18).

18 Regret: Measuring Algorithm Performance

18.1 Definition of Regret

To evaluate the performance of a bandit algorithm, we use the notion of **regret**—the total loss incurred by not always playing the optimal arm.

Let $\alpha^* = \max\{\alpha_0, \alpha_1, \ldots, \alpha_J\}$ denote the true success probability of the best arm. Without loss of generality (WLOG), assume arm $j = 0$ is the best arm, so $\alpha^* = \alpha_0$.

Definition 18.1 (Cumulative Regret)**.** After T rounds, the cumulative regret is:

$$R(T) = \sum_{j=1}^{J} N_j(T) \cdot (\alpha_0 - \alpha_j)$$

where $N_j(T)$ is the total number of times arm j was pulled over T rounds, and $(\alpha_0 - \alpha_j) \geq 0$ is the **suboptimality gap** of arm j.

Intuitively, $R(T)$ counts the number of potential losses from pulling suboptimal arms: each time we pull arm $j \neq 0$, we lose $(\alpha_0 - \alpha_j)$ in expectation compared to pulling the best arm.

The **expected regret** is:

$$\mathbb{E}[R(T)] = \sum_{j=1}^{J} \mathbb{E}[N_j(T)] \cdot (\alpha_0 - \alpha_j)$$

This is the standard criterion used to evaluate the optimality of a bandit algorithm.

18.2 Regret Bound for Thompson Sampling

Definition 18.2 (Kullback–Leibler Divergence for Bernoulli)**.** The KL divergence between two Bernoulli distributions with success probabilities q and p is:

$$\mathbb{D}_{\mathrm{KL}}(q \,\|\, p) = q \ln\left(\frac{q}{p}\right) + (1-q) \ln\left(\frac{1-q}{1-p}\right)$$

This measures the "distance" between two Bernoulli distributions. Note $\mathbb{D}_{\mathrm{KL}}(q\|p) \geq 0$ with equality iff $p = q$.

The KL divergence plays a central role in the regret bound because it captures how hard it is to distinguish arm j from the best arm based on observed data.

Regret Bound: Agrawal & Goyal (2017)

For every $\alpha_0, \alpha_1, \ldots, \alpha_J$ where WLOG $\alpha_0 = \max(\alpha_0, \ldots, \alpha_J)$, there exists a constant

C such that for all $\varepsilon > 0$:

$$\mathbb{E}[R(T)] \;\leq\; (1+\varepsilon)^2 \sum_{j=1}^{J} \frac{\ln T \cdot (\alpha_0 - \alpha_j)}{\mathbb{D}_{\mathrm{KL}}(\alpha_j \,\|\, \alpha_0)} + \frac{C \cdot J}{\varepsilon^2}$$

Interpretation:

- This is an **upper bound** on total regret.
- The dominant term grows as $O(\ln T)$, meaning total regret is *logarithmic* in T.
- Thus $\mathbb{E}[R(T)] = O(\ln T)$, which implies expected regret *per round* is:

$$\frac{\mathbb{E}[R(T)]}{T} = O\left(\frac{\ln T}{T}\right) \to 0 \text{ as } T \to \infty.$$

- Thompson Sampling **eventually uses the best arm most of the time**.

When is Regret Larger? Regret tends to be larger when:

1. The arm success probabilities are all similar: $\alpha_0 \approx \alpha_1 \approx \cdots \approx \alpha_J$. In this case $\mathbb{D}_{\mathrm{KL}}(\alpha_j \| \alpha_0) \approx 0$, making the ratio blow up. This makes sense intuitively: if all arms are nearly identical, it takes a lot of data to identify the best one.
2. **Arms might be similar in structure**—if learning about one arm teaches you about another (e.g., correlated arms), the standard bandit model can be extended to exploit this.

18.3 Extensions

- **Contextual Bandits:** In the standard bandit, the agent chooses an arm with no additional side information. In contextual bandits, each round presents a *context* vector x_t (individual-level features), and the reward distribution for arm j depends on x_t. The agent must learn a policy $\pi(x_t) \to j$ mapping contexts to arms.
- **Correlated arms:** If arms are structurally similar (e.g., different prices for the same product), learning about one updates beliefs about others, enabling faster learning.

19 Combinatorial Optimization and UPS-ORION

19.1 The Vehicle Routing Problem

UPS-ORION (On-Road Integrated Optimization and Navigation) is a real-world combinatorial optimization system. The underlying problem is a variant of the **Traveling Salesman / Vehicle Routing Problem**.

Setup: Consider a set of locations (nodes/vertices) representing intersections or delivery stops. The graph notation is:

- $i, j, \ldots$: **vertices/nodes** (intersections or delivery locations)
- ij: **arc** (directed edge from i to j)
- x_{ijt}: **travel time** from i to j at trip t (a random variable)

19.2 The Exploration-Exploitation Connection in Routing

The routing problem is a bandit problem in disguise. Each arc ij is an "arm" with an unknown mean travel time $\mu_{ij} = \mathbb{E}[X_{ijt}]$. The algorithm must balance:

- **Exploiting** known fast routes (take the route with the best estimated travel time).
- **Exploring** alternative routes that haven't been driven recently (to update travel time estimates).

19.3 Tracking Travel Time History

Let $D_{ijs} = \mathbb{1}[\text{arc } ij \text{ was driven on trip } s]$. Then:

$$\bar{n}_{ijt} = \sum_{s=0}^{t-1} D_{ijs} \qquad (\text{\# of times arc } ij \text{ was driven in prior trips})$$

$$\bar{X}_{ijt} = \frac{1}{\bar{n}_{ijt}} \sum_{s=0}^{t-1} D_{ijs} \cdot X_{ijs} \qquad (\text{average travel time for arc } ij \text{ over past trips})$$

19.4 UCB-Style Cost Function

The algorithm uses an **Upper Confidence Bound (UCB)**-style cost for each arc:

$$C_{ijt} = \bar{X}_{ijt} - \sqrt{\frac{2 \ln t}{\bar{n}_{ijt}}}$$

- The first term $\bar{X}_{ijt}$ is the empirical mean travel time (lower is better, since we minimize travel time).
- The second term $-\sqrt{2 \ln t / \bar{n}_{ijt}}$ is a **bonus for exploration**: it makes under-explored arcs (small $\bar{n}_{ijt}$) appear cheaper (more attractive), incentivizing the driver to try them.
- As $\bar{n}_{ijt} \to \infty$, the exploration bonus $\to 0$, and the cost converges to the true mean travel time.
- If an arc has bad luck (high realized travel time on a few trips), it gets a large exploration bonus, encouraging future exploration of that arc.

19.5 The ORION Recipe

At each trip t:

1. Use C_{ijt} as the edge weight for all arcs ij.
2. Compute the **minimum-cost route** (solving a shortest-path or TSP subproblem) using these costs.
3. Execute the route, recording the actual time X_{ijs} spent on each segment.
4. Update $\bar{X}_{ijt}$ and $\bar{n}_{ijt}$ for each arc driven.

19.6 Hoeffding's Inequality Justification

Why does $\sqrt{2\ln t/\bar{n}_{ijt}}$ arise naturally? The answer comes from **Hoeffding's Inequality**.

Theorem 19.1 (Hoeffding's Inequality)**.** *Let $X_{ij1}, \ldots, X_{ijn}$ be i.i.d. bounded random variables with mean μ_{ij}. Then:*

$$\Pr\big(\big|\bar{X}_{ijt} - \mu_{ij}\big| \geq \delta\big) \leq 2\exp(-2\bar{n}_{ijt}\,\delta^2)$$

Interpretation: The probability that our sample mean deviates from the true mean by more than δ is bounded exponentially in the number of observations $\bar{n}_{ijt}$. More data $\Rightarrow$ less chance of being wrong by more than δ.

Deriving δ: Set the bound equal to $2/t^4$ (a common choice ensuring a union bound over t and arcs):

$$\Pr\big(\big|\bar{X}_{ijt} - \mu_{ij}\big| > \delta\big) \leq \frac{2}{t^4} < 2\exp\big(-2\bar{n}_{ijt}\,\delta^2\big)$$

Setting these equal and solving:

$$\delta = \sqrt{\frac{2\ln t}{\bar{n}_{ijt}}}$$

Confidence Guarantee

With ex-ante probability at least $1 - 2/t^4$, the sample average $\bar{X}_{ijt}$ will be no further than $\delta = \sqrt{2\ln t/\bar{n}_{ijt}}$ from the true mean μ_{ij}.

This is precisely the exploration bonus in the ORION cost function: it represents the uncertainty radius around the mean travel time estimate, with high probability.

20 MCMC Cipher Decoding

20.1 Historical Motivation: Alan Turing and the Enigma

Alan Turing famously used early MCMC-like methods to crack the German Enigma cipher in World War II. The core idea—now formalized as MCMC—is to explore a large, intractable space (all possible cipher keys) by making local random proposals and accepting or rejecting them based on a likelihood ratio.

20.2 Cipher Key Setup

Definition 20.1 (Cipher Key)**.** Let $\mathbb{A}$ be the **plaintext alphabet** $= \{a, b, c, \ldots, z\}$ (with $|\mathbb{A}| = J$) and $\mathbb{B}$ be the **ciphertext alphabet** $= \{\alpha, \beta, \gamma, \ldots\}$ with $|\mathbb{B}| = |\mathbb{A}|$.

A **cipher key** is a bijection $\phi : \mathbb{B} \to \mathbb{A}$ (mapping ciphertext symbols to plaintext symbols). There are $L = J!$ possible cipher keys.

Example: With $\mathbb{A} = \{a, b, c\}$ and $\mathbb{B} = \{\alpha, \beta, \gamma\}$, one cipher key is:

$$\phi = \{(\alpha, a),\ (\beta, b),\ (\gamma, c)\}$$

meaning $\phi(\alpha) = a$, $\phi(\beta) = b$, $\phi(\gamma) = c$.

If we observe ciphertext $y = \alpha\alpha\gamma\beta\alpha\gamma$, then:

$$\phi(y) = \tilde{y} = aacbac \quad \text{(decrypted plaintext)}$$

and the inverse $\phi^{-1}(\tilde{y}) = y$ re-encrypts the plaintext.

20.3 Probability Model for Plaintexts

We assume a **population of English plaintexts** exists. Let:

- $\tilde{Y}$: a random draw from the population of all plain texts.
- $\tilde{y}$: the specific plaintext we actually observe (after decryption).

The **likelihood** of observing decrypted text $\tilde{y}$ given cipher ϕ is:

$$\Pr(\tilde{Y} = \tilde{y} \mid \phi) = \Pr\big(\phi^{-1}(Y) = y\big)$$

where Y is the observed ciphertext. That is, we compute the probability that a random English text, when encrypted with cipher ϕ, produces our observed ciphertext y.

20.4 Composite Likelihood via Bigrams

Directly modeling arbitrary-length texts is intractable. Instead, we use a **bigram model** (a simplifying approximation):

Definition 20.2 (Bigram)**.** A **bigram** is a pair of consecutive letters in a text. For example, "Peter piper picked a peck of pickled peppers" begins with bigrams: "pe", "et", "te", "er", "r ", ...

Let $\pi_{aa'}$ denote the **population frequency** of bigram aa' (two consecutive letters $a, a' \in \mathbb{A}$) in English plaintexts. These frequencies can be estimated from a large English corpus (e.g., the collected works of Shakespeare, or a newspaper archive). Note $\sum_{a,a' \in \mathbb{A}} \pi_{aa'} = 1$.

Bigram likelihood: For a single randomly sampled bigram $\hat{Y} \in \mathbb{A} \times \mathbb{A}$:

$$\Pr(\hat{Y} = \hat{y}) = \prod_{a \in \mathbb{A}} \prod_{a' \in \mathbb{A}} \pi_{aa'}^{\mathbb{1}(\hat{y} = aa')}$$

For a ciphertext y of length N bigrams, with k-th bigram y_k, the **composite likelihood** under cipher ϕ is:

Composite Likelihood

$$\ell_N^{cl}(\phi) = \prod_{k=1}^{N} \left\{ \prod_{a \in \mathbb{A}} \prod_{a' \in \mathbb{A}} \pi_{aa'}^{\mathbb{1}(\phi(y_k) = aa')} \right\}$$

Interpretation:

- For each bigram y_k in the ciphertext, apply cipher ϕ to decode it, getting decrypted bigram $\phi(y_k)$.
- Look up the English frequency $\pi_{\phi(y_k)}$ for that decoded bigram.
- The composite likelihood is the product of all these frequencies across all N bigrams.
- **Higher likelihood** $\Rightarrow$ the decrypted text looks more like natural English.
- This is an *incorrect* (misspecified) model because consecutive bigrams are treated as independent, but it works well in practice because it captures the most important first-order structure of the language.

Log-likelihood: In practice, we work with the **log composite likelihood** to avoid numerical underflow:

$$\ln \ell_N^{cl}(\phi) = \sum_{k=1}^{N} \sum_{a \in \mathbb{A}} \sum_{a' \in \mathbb{A}} \mathbb{1}(\phi(y_k) = aa') \cdot \ln \pi_{aa'}$$

20.5 Bayesian Posterior over Cipher Keys

We place a **uniform prior** over all $L = J!$ possible ciphers:

$$\Pr(\phi_0 = \phi) = \Pi(\phi) = \frac{1}{J!} = \frac{1}{L}$$

Using Bayes' Rule, the posterior probability of cipher ϕ given observed ciphertext y is:

$$\pi(\phi \mid y) = \frac{\Pr(y \mid \phi)\,\Pi(\phi)}{\Pr(y)} = \frac{\Pr(y \mid \phi) \cdot \frac{1}{L}}{\sum_{\ell=1}^{L} \Pr(y \mid \phi_\ell) \cdot \frac{1}{L}} = \frac{\Pr(y \mid \phi)}{\sum_{\ell=1}^{L} \Pr(y \mid \phi_\ell)}$$

Replacing the exact likelihoods with composite likelihoods:

$$\pi^{cp}(\phi \mid y) \approx \frac{\ell_N^{cl}(\phi)}{\sum_{\ell=1}^{L} \ell_N^{cl}(\phi_\ell)}$$

The problem: $L = J!$ is astronomically large ($26! \approx 4 \times 10^{26}$ for the English alphabet). We cannot compute this sum directly. This is where MCMC comes in.

21 Markov Chain Monte Carlo (MCMC): Metropolis-Hastings

21.1 Motivation

Since we cannot enumerate all $J!$ cipher keys, we need a smarter approach. MCMC provides a way to **sample from the posterior** $\pi^{cp}(\phi \mid y)$ without ever computing the normalizing constant $\sum_\ell \ell_N^{cl}(\phi_\ell)$.

21.2 The Metropolis-Hastings Algorithm for Cipher Decoding

MCMC Cipher Decoding (Metropolis-Hastings)

Input: Ciphertext y, bigram frequencies $\{\pi_{aa'}\}$, number of steps S.

1. **Initialize:** Pick an initial cipher $\phi^{(0)} \in \Phi$ (e.g., at random or using frequency analysis). Set $s = 0$.
2. **Propose:** Construct a proposed cipher $\tilde{\phi}^{(s+1)}$ by randomly **transposing two mappings** in $\phi^{(s)}$:
   - Pick two distinct ciphertext letters $c, c' \in \mathbb{B}$ uniformly at random.
   - Swap their decryptions: if $\phi^{(s)}(c) = z$ and $\phi^{(s)}(c') = q$, set $\tilde{\phi}^{(s+1)}(c) = q$ and $\tilde{\phi}^{(s+1)}(c') = z$.
3. **Accept/Reject:** Compute the **acceptance ratio** (ratio of composite posteri-

ors):

$$r = \frac{\pi^{cp}(\tilde{\phi}^{(s+1)} \mid y)}{\pi^{cp}(\phi^{(s)} \mid y)} = \frac{\ell_N^{cl}(\tilde{\phi}^{(s+1)})}{\ell_N^{cl}(\phi^{(s)})}$$

Note: the normalizing constants cancel!

- If $r \geq 1$ (proposed key has higher posterior): set $\phi^{(s+1)} = \tilde{\phi}^{(s+1)}$.
- If $r < 1$: set $\phi^{(s+1)} = \tilde{\phi}^{(s+1)}$ with probability r, otherwise $\phi^{(s+1)} = \phi^{(s)}$.

4. Set $s \leftarrow s + 1$ and return to Step 2 until $s = S$.

21.3 Key Properties

Why the normalizing constant cancels:

$$\frac{\pi^{cp}(\tilde{\phi} \mid y)}{\pi^{cp}(\phi \mid y)} = \frac{\ell_N^{cl}(\tilde{\phi}) / \sum_\ell \ell_N^{cl}(\phi_\ell)}{\ell_N^{cl}(\phi) / \sum_\ell \ell_N^{cl}(\phi_\ell)} = \frac{\ell_N^{cl}(\tilde{\phi})}{\ell_N^{cl}(\phi)}$$

It is just a ratio of numerators; the denominators cancel out. This is the core computational trick.

Exploration-Exploitation Analogy: The proposal step $\tilde{\phi}^{(s+1)}$ plays the same role as the exploration mechanism in Thompson Sampling:

- It randomly **explores** nearby ciphers (swapping 2 mappings = small local change).
- The accept/reject step ensures the chain spends more time near high-posterior regions (**exploitation**).

Markov Property: The sequence $\phi^{(0)}, \phi^{(1)}, \ldots, \phi^{(S)}$ is a **Markov chain**: the next state depends only on the current state (not the full history).

Convergence: Under mild conditions, the chain converges to the stationary distribution $\pi^{cp}(\phi \mid y)$. As $S \to \infty$, the empirical distribution of $\{\phi^{(s)}\}$ approximates the true posterior. In practice, the posterior is highly concentrated on a few keys (or just one—the true cipher), as illustrated by the spike in the distribution diagram.

Accept-Reject in Plain Terms:

- **If the new key is better** (higher likelihood, meaning the decoded text looks more like English), always keep it.
- **If the new key is worse**, sometimes still switch to it with probability $r < 1$. This prevents getting stuck in local optima and ensures the chain can escape suboptimal configurations.

Practical Notes:

- It is typically not efficient to just randomly try $\phi_i \in \Phi$ exhaustively; an efficient algorithm also builds on its previous state (hence the Markov chain structure).
- We use the **log likelihood** in practice:

$$\ln r = \ln \ell_N^{cl}(\tilde{\phi}) - \ln \ell_N^{cl}(\phi^{(s)})$$

  to avoid numerical underflow from multiplying many small probabilities.
- The proposal distribution (swapping 2 letters) is symmetric: $q(\tilde{\phi} \mid \phi) = q(\phi \mid \tilde{\phi})$, which is why the acceptance probability simplifies to just the likelihood ratio (the Metropolis, not Metropolis-Hastings, case).

22 Sequential Likelihoods and the Hot Hand

22.1 Sequential / Recursive Factoring

Consider a sequence of binary outcomes $Y_0, Y_1, \ldots, Y_T \in \{0, 1\}$ (e.g., Steph Curry making or missing consecutive 3-pointers).

The joint probability can be factored sequentially:

$$P(Y^{0:T}) = P(Y_T \mid Y^{0:T-1}) \cdot P(Y^{0:T-1})$$

Applying this recursively:

$$P(Y^{0:T}) = P(Y_T \mid Y^{0:T-1}) \cdot P(Y_{T-1} \mid Y^{0:T-2}) \cdots P(Y_1 \mid Y_0) \cdot P(Y_0)$$

This is the exact **likelihood** of the full sequence, making no independence assumptions.

22.2 The Independence Approximation

If we assume outcomes are **independent** (ignoring temporal dependence):

$$P(Y^{0:T}) \approx P(Y^T) \cdots P(Y^0) = \prod_{t=0}^{T} P(Y_t)$$

Taking logs:

$$\ln P(Y^{0:T}) \approx \sum_{t=0}^{T} \ln P(Y_t)$$

This approximation *ignores the fact that events have a dependence on each other.* It allows for **modularity**: each period's probability can be estimated or modeled separately.

22.3 The Hot Hand

The **Hot Hand** phenomenon is the empirical claim that in basketball (and other settings), a player who has just succeeded is more likely to succeed again:

$$\Pr(Y_1 = 1 \mid Y_0 = 1) > \Pr(Y_1 = 1 \mid Y_0 = 0)$$

If the hot hand is real, then the sequential factoring model (which allows dependence) is correct, while the independence approximation is misspecified. The statistical challenge is distinguishing genuine hot-hand effects from random clustering that occurs by chance even when shots are independent (the *hot hand fallacy*).

In the context of cipher decoding, the bigram composite likelihood is essentially an independence approximation (treating each bigram as an independent draw) that ignores higher-order dependencies in language.

23 Summary Table

Concept	**Formula**	**Interpretation**
Posterior	$\alpha_j \mid \mathcal{I}_{t-1} \sim \text{Beta}(S_j + a_j,\, F_j + b_j)$	Updated belief after observing data
Posterior Mean	$\mathbb{E}[\alpha_j \mid \mathcal{I}_{t-1}] = \frac{S_j + a_j}{N_j + a_j + b_j}$	Approx. success rate
Regret	$R(T) = \sum_{j=1}^{J} N_j(T)(\alpha_0 - \alpha_j)$	Cumulative loss vs. optimal
TS Bound	$\mathbb{E}[R(T)] \leq (1 + \varepsilon)^2 \sum_j \frac{\ln T \cdot \Delta_j}{\mathbb{D}_{KL}(\alpha_j \Vert \alpha_0)} + \frac{CJ}{\varepsilon^2}$	Logarithmic regret
ORION Cost	$C_{ijt} = \bar{X}_{ijt} - \sqrt{2 \ln t / \bar{n}_{ijt}}$	UCB for route arcs
Hoeffding	$\Pr(\lvert \bar{X} - \mu \rvert \geq \delta) \leq 2e^{-2n\delta^2}$	Concentration of mean
Composite Lik.	$\ell_N^{cl}(\phi) = \prod_{k=1}^{N} \prod_{a,a'} \pi_{aa'}^{\mathbb{1}(\phi(y_k)=aa')}$	Bigram likelihood under cipher
MCMC Ratio	$r = \ell_N^{cl}(\tilde{\phi}) / \ell_N^{cl}(\phi^{(s)})$	Accept/reject proposal

Linear Regression: BLP, OVB, Frisch-Waugh, and Card-Krueger

24 Linear Regression Redux

24.1 The Setup: Population Joint Distribution

We are interested in the joint distribution of two variables in some population:

$$(X, Y) \sim F_{X,Y}$$

In the running example:

- X = years of schooling
- Y = log earnings

$F_{X,Y}$ is the joint distribution of schooling–earnings pairs in the population. Think of it as a large table summarizing the frequency of every (x, y) combination. In practice, $F_{X,Y}$ is unknown but **asymptotically revealed under random sampling**: as we collect more data, our empirical joint distribution converges to the true $F_{X,Y}$.

Two Types of Analysis:

1. **Infinite-data / population analysis:** What would we conclude if we had perfect knowledge of $F_{X,Y}$? This is the object of population linear regression.
2. **Finite-sample analysis:** How do we correct for errors that arise because we only have a finite sample?

24.2 The Prediction Problem

Motivating example: You move to a new neighborhood. Your neighbor reveals she has X years of schooling. What is your best guess of her log earnings Y?

We want to find a linear predictor of the form $a + bX$ that minimizes expected prediction error. The natural loss function is **squared error loss**:

Definition 24.1 (Squared Error Loss and Risk)**.** Given a linear predictor $a + bX$, the **prediction error** (residual) is:

$$U(a, b) = Y - a - bX$$

The **loss** (negative utility) from an incorrect prediction is:

$$U(a, b)^2 = (Y - a - bX)^2$$

The **risk** (expected loss) is:

$$\mathcal{R}(a, b) = \mathbb{E}\big[(Y - a - bX)^2\big]$$

The squared-error loss function is a symmetric bowl shape: it penalizes *both* over-prediction and under-prediction symmetrically and increasingly, placing more weight on large errors.

Remark 24.2. **Why squared error?** Because it is analytically tractable: we can take derivatives using the power rule and obtain closed-form solutions. It is the *mean squared error (MSE)-minimizing* criterion. Other loss functions (e.g., absolute error) would give different population objects (e.g., the conditional median).

Minimizing $\mathcal{R}(a, b)$ over $(a, b) \in \mathbb{R}^2$ is operationally feasible because we can compute expectations under $F_{X,Y}$.

In the continuous case:

$$\mathbb{E}\big[(Y - a - bX)^2\big] = \iint (y - a - bx)^2 \, f_{X,Y}(x, y) \, dx \, dy$$

In the discrete case:

$$\mathbb{E}\big[(Y - a - bX)^2\big] = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} (y - a - bx)^2 \, f_{X,Y}(x, y)$$

where $f_{X,Y}(x, y)$ is the PMF (probability mass function) in the discrete case.

25 First-Order Conditions and the BLP Solution

25.1 Solving the MSE Problem

We minimize $\mathcal{R}(a, b)$ by differentiating with respect to a and b and setting to zero.

First-Order Conditions (FOCs):

$$\frac{\partial \mathcal{R}}{\partial a} = 0 \implies \mathbb{E}[Y - \alpha_0 - \beta_0 X] = \mathbb{E}[U] = 0 \tag{1}$$

$$\frac{\partial \mathcal{R}}{\partial b} = 0 \implies \mathbb{E}[(Y - \alpha_0 - \beta_0 X) \cdot X] = \mathbb{E}[UX] = 0 \tag{2}$$

where α_0, β_0 are the optimal (MSE-minimizing) intercept and slope.

Population BLP Solution

Solving the two FOCs simultaneously yields:

$$\beta_0 \stackrel{\text{def}}{=} \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}, \qquad \alpha_0 \stackrel{\text{def}}{=} \mathbb{E}[Y] - \mathbb{E}[X] \, \beta_0$$

The optimal prediction error is:

$$U \;=\; Y - \alpha_0 - \beta_0 X$$

satisfying $\mathbb{E}[U] = 0$ and $\mathbb{E}[UX] = 0$ (equivalently, $\mathrm{Cov}(U, X) = 0$).

25.2 Key Properties of the BLP

Definition 25.1 (Best Linear Predictor (BLP))**.** The **Best Linear Predictor** of Y given X is:

$$E^*[Y \mid X] \stackrel{\text{def}}{=} \alpha_0 + \beta_0 X$$

This is notation for the MSE-minimizing linear predictor of Y given X under the population distribution $F_{X,Y}$.

Remark 25.2 (BLP vs. CEF). The BLP $E^*[Y \mid X]$ and the **Conditional Expectation Function** (CEF) $\mathbb{E}[Y \mid X]$ are **not the same in general**:

- $\mathbb{E}[Y \mid X]$ is the true conditional mean of Y given $X = x$. It can be any function of x.
- $E^*[Y \mid X] = \alpha_0 + \beta_0 X$ is the best *linear* approximation to $\mathbb{E}[Y \mid X]$.

They coincide only when the true CEF is linear (e.g., under joint normality of (X, Y)).

Orthogonality conditions: The two FOCs are equivalent to:

$$\mathbb{E}[UX] = 0 \quad \Longleftrightarrow \quad \text{Cov}(U, X) = 0$$

because $\mathbb{E}[UX] - \mathbb{E}[U]\mathbb{E}[X] = \text{Cov}(U, X)$ and $\mathbb{E}[U] = 0$, so $\mathbb{E}[UX] = \text{Cov}(U, X)$.

Geometrically: the BLP is the linear projection of Y onto the space spanned by $(1, X)$.

Sample analogues: With finite data $\{(x_i, y_i)\}_{i=1}^n$, the sample OLS estimators are:

$$\hat{\beta} = \frac{S_{XY}}{S_X^2}, \qquad \hat{\alpha} = \bar{Y} - \bar{X}\hat{\beta}$$

where $S_{XY} = \frac{1}{n}\sum_i (x_i - \bar{x})(y_i - \bar{y})$ is the sample covariance and S_X^2 is the sample variance of X.

26 Short, Long, and Auxiliary Regressions (Goldberger)

26.1 Setup and Definitions

Following **Arthur Goldberger**, consider three regressions involving outcome Y, main regressor X, and a third variable W:

Definition 26.1 (Long, Short, and Auxiliary Regressions)**.**

$$\textbf{(LR) Long Regression:} \quad E^*[Y \mid X, W] = \alpha_0 + \beta_0 X + \gamma_0 W \tag{3}$$

$$\textbf{(SR) Short Regression:} \quad E^*[Y \mid X] = a_0 + b_0 X \tag{4}$$

$$\textbf{(AR) Auxiliary Regression:} \quad E^*[W \mid X] = \phi_0 + \pi_0 X \tag{5}$$

In the schooling context: Y = log earnings, X = years of schooling, W = AFQT score (Armed Forces Qualification Test, a measure of cognitive ability).

Note that $\pi_0 > 0$ means that individuals with more years of schooling tend to have higher AFQT scores—ability and schooling are positively correlated in the population.

26.2 The Omitted Variable Bias (OVB) Formula

A fundamental result in regression analysis is how the short regression coefficient b_0 relates to the long regression coefficient β_0.

Derivation via the Law of Iterated Linear Predictors (LILP): The long regression model is:

$$Y = \alpha_0 + \beta_0 X + \gamma_0 W + U, \quad \text{where } \mathbb{E}[UX] = \mathbb{E}[UW] = \mathbb{E}[U] = 0.$$

Using the fact that **linear regression is a linear operator** ($E^*[X + Y \mid \cdot] = E^*[X \mid \cdot] + E^*[Y \mid \cdot]$):

$$\begin{aligned} E^*[Y \mid X] &= \alpha_0 + \beta_0 X + \gamma_0 \underbrace{E^*[W \mid X]}_{\text{from AR}} + E^*[U \mid X] \\ &= \alpha_0 + \beta_0 X + \gamma_0(\phi_0 + \pi_0 X) + 0 \\ &= (\alpha_0 + \gamma_0\phi_0) + (\beta_0 + \gamma_0\pi_0)X \end{aligned}$$

Comparing with $E^*[Y \mid X] = a_0 + b_0 X$, we read off:

Omitted Variable Bias (OVB) Formula

$$b_0 = \beta_0 + \gamma_0 \cdot \pi_0$$

Equivalently, rearranging:

$$\underbrace{b_0}_{\text{short}} - \underbrace{\beta_0}_{\text{long}} = \underbrace{\gamma_0}_{\substack{\text{effect of } W \\ \text{on } Y}} \cdot \underbrace{\pi_0}_{\substack{\text{effect of } X \\ \text{on } W}}$$

b_0 is the **short regression coefficient** (omitting W); β_0 is the **long regression coefficient** (including W).

- $b_0 = \beta_0$ (no omitted variable bias) if and only if $\gamma_0 = 0$ (W has no effect on Y) *or* $\pi_0 = 0$ (W is uncorrelated with X).
- $b_0 > \beta_0$ (upward bias) if $\gamma_0\pi_0 > 0$: either both positive (ability raises earnings *and* is correlated with schooling) or both negative.
- $b_0 < \beta_0$ (downward bias) if $\gamma_0\pi_0 < 0$.

Example 26.2 (Marriage as an Omitted Variable)**.** If W = marital status: marriage is correlated with higher earnings ($\gamma_0 > 0$) and is more common among people with more schooling ($\pi_0 > 0$). Including or excluding marriage generates OVB in either direction depending on the direction of correlation.

26.3 The Law of Iterated Expectations (LIE) for Linear Predictors

A key tool used above is:

$$E^*[Y \mid X] = E^*[E^*[Y \mid X, W] \mid X]$$

This is the **Law of Iterated Linear Projections (LILP)**, the analogue of the Law of Iterated Expectations for linear projections. It connects the short and long regression functions: you can always compute the short regression by "averaging out" W from the long regression.

27 Residual Regression and the Frisch–Waugh Theorem

27.1 The Residual V

Definition 27.1 (Residual from Auxiliary Regression)**.** Define V as the **variation in W that is linearly unpredictable from X**:

$$V \stackrel{\text{def}}{=} W - \phi_0 - \pi_0 X = W - E^*[W \mid X]$$

So $W = \phi_0 + \pi_0 X + V$, and by construction $\mathrm{Cov}(V, X) = 0$ and $\mathbb{E}[V] = 0$.

V is the "X-partialled-out" component of W: it captures the idiosyncratic variation in ability that is not linearly predictable from schooling.

27.2 Proto-Residual Regression

Substituting $W = \phi_0 + \pi_0 X + V$ into the long regression:

$$\begin{aligned} E^*[Y \mid X, W] &= \alpha_0 + \beta_0 X + \gamma_0(\phi_0 + \pi_0 X + V) \\ &= (\alpha_0 + \gamma_0\phi_0) + (\beta_0 + \gamma_0\pi_0)X + \gamma_0 V \\ &= E^*[Y \mid X, V] \end{aligned}$$

Key Insight: Rewriting in Terms of Residuals

Any linear function of X and W can be rewritten as a linear function of X and V:

$$E^*[Y \mid X, V] = (\alpha_0 + \gamma_0\phi_0) + (\beta_0 + \gamma_0\pi_0)\, X + \gamma_0 V$$

The coefficient on V is γ_0—exactly the same as the coefficient on W in the long regression. This makes sense because V is the "new information" in W beyond what X already tells us.

27.3 The Frisch–Waugh Theorem

Now apply the Law of Iterated Linear Projections (LILP): project $E^*[Y \mid X, V]$ onto V alone, using the fact that $\mathrm{Cov}(V, X) = 0$ (so $E^*[X \mid V] = 0$ after demeaning):

Frisch–Waugh (Residual Regression) Theorem

$$E^*[Y \mid V] = (\alpha_0 + \gamma_0\phi_0) + \gamma_0 V$$

The coefficient on V in the regression of Y on V alone equals γ_0, the coefficient on W in the long regression of Y on (X, W).

Statement of the theorem: The Frisch–Waugh theorem says that the coefficient γ_0 on W in the multiple regression of Y on (X, W) can equivalently be obtained by:

1. Regress Y on X; save residuals $\tilde{Y} = Y - E^*[Y \mid X]$.
2. Regress W on X; save residuals $V = W - E^*[W \mid X]$.
3. Regress $\tilde{Y}$ on V.

The slope from step 3 equals γ_0.

Intuition: By partialling out X from both Y and W first, we remove the shared variation driven by X, and isolate the "pure" relationship between W and Y that is orthogonal to X.

28 The Card–Krueger (1996) Model of Schooling

28.1 The Structural Model

Card and Krueger (1994/1996) develop a structural model of schooling and earnings to understand what regression coefficients actually measure. The model is built around a **potential outcomes / response function** framework.

Definition 28.1 (Potential Earnings Response Function)**.** For an agent with characteristics (A, C, V^*), if they attain x years of schooling, their log earnings would be:

$$\log Y(x) = \beta_0 x + A + V^*$$

where:

- β_0 = common **return to schooling** (semi-elasticity: $\frac{d \log Y(x)}{dx} \cdot \frac{1}{Y(x)} = \beta_0$ percent increase in earnings per additional year)
- A = individual **ability** (unobserved systematic driver of earnings differences)
- V^* = idiosyncratic **luck** (unsystematic earnings shocks)
- $x \in \mathcal{X}$ = years of schooling (the choice variable)

28.2 The Cost of Schooling

The cost of obtaining x years of schooling is quadratic in x with agent-specific intercept C:

$$Z(x) = C\,x + \frac{k_0}{2}x^2$$

where:

- C = agent-specific marginal cost of schooling at $x = 0$ (heterogeneous across agents)
- $k_0 > 0$ = common curvature parameter (marginal cost is increasing in x: $dZ/dx = C + k_0 x$)

The marginal cost of an additional year of schooling is $C + k_0 x$, which is increasing in x (diminishing returns to human capital investment at the margin).

28.3 Heterogeneity Distribution

Agent heterogeneity is fully captured by (A, C, V^*):

- $(A, C) \sim F_{A,C}$: joint distribution of ability and cost, potentially correlated
- $V^* \sim F_{V^*}$: luck, drawn independently of (A, C), i.e., $(A, C) \perp\!\!\!\perp V^*$

The full distributional assumptions are:

$$\begin{pmatrix} A \\ C \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_A \\ \mu_C \end{pmatrix}, \begin{pmatrix} \sigma_A^2 & \sigma_{AC} \\ \sigma_{AC} & \sigma_C^2 \end{pmatrix}\right)$$

$$V^*(A, C) \sim N(\alpha_0, \sigma_v^2) \quad \text{(conditional on any } (A, C)\text{, since independent)}$$

- σ_{AC} = covariance between ability and cost: this is **crucial** for understanding bias.
- Option 1 ("intuitive"): $C_H \Rightarrow A_H$ and $C_L \Rightarrow A_L$ — high-cost individuals have high ability. This gives $\sigma_{AC} > 0$.
- Option 2 ("ability bias" scenario, assumed to be true for the argument): $C_H \Rightarrow A_L$ and $C_L \Rightarrow A_H$ — high-cost individuals have *low* ability. This gives $\sigma_{AC} < 0$.

28.4 The Schooling Decision Problem

An agent's **utility** from choosing x years of schooling is earnings minus costs:

$$\text{Utility}(x) = \log Y(x) - Z(x) = (\beta_0 x + A + V^*) - \left(Cx + \frac{k_0}{2}x^2\right)$$

The agent's information set: they know how the world works, know their own (A, C), but **cannot predict luck** V^*.

The optimal schooling choice X^* solves:

$$X^* = \arg\max_{x \in \mathcal{X}} \mathbb{E}[\log Y(x) - Z(x) \mid A, C]$$

Since $\mathbb{E}[V^* \mid A, C] = \alpha_0$ (a constant, because $V^* \perp (A, C)$), this simplifies to:

$$X^* = \arg\max_{x \in \mathcal{X}} \left(\beta_0 x + A + \alpha_0 - Cx - \frac{k_0}{2} x^2 \right)$$

Taking the first-order condition $\frac{d}{dx}[\cdot] = 0$:

$$\beta_0 - C - k_0 x = 0$$

Optimal Schooling Choice

$$X^* = \frac{1}{k_0}(\beta_0 - C)$$

The agent chooses more schooling when:

- β_0 is higher (greater return to schooling)
- C is lower (lower marginal cost)
- k_0 is lower (less convexity in costs / less rapidly increasing marginal cost)

Graphical interpretation: The marginal benefit of schooling is $\frac{d \log Y}{dx} = \beta_0$ (constant, horizontal line). The marginal cost is $C + k_0 x$ (upward-sloping). The agent sets MB = MC, giving $X^* = (\beta_0 - C)/k_0$.

29 Ability Bias: What Does the Short Regression Measure?

29.1 Computing the Short Regression Coefficient

In data, we observe $(X, Y) = (X^*, \log Y(X^*))$. The observed log earnings are:

$$\log Y = \beta_0 X + A + V$$

where $V \stackrel{\text{def}}{=} V^* - \alpha_0$ has $\mathbb{E}[V \mid A, C] = 0$ and $V \perp (A, C)$.

The short regression of $\log Y$ on X gives slope:

$$b_0 = \frac{\text{Cov}(X, \log Y)}{\text{Var}(X)}$$

We need to compute both $\text{Cov}(X, \log Y)$ and $\text{Var}(X)$.

Step 1: Moments of X. Recall $X = \frac{1}{k_0}(\beta_0 - C)$. Since β_0 and k_0 are constants:

$$\mathbb{E}[X] = \frac{1}{k_0}(\beta_0 - \mu_C), \qquad \text{Var}(X) = \text{Var}\left(\frac{1}{k_0}C\right) = \frac{\sigma_C^2}{k_0^2}$$

(The variance of β_0/k_0 is zero since it's a constant.)

Step 2: Mean of $\log Y$.

$$\mathbb{E}[\log Y] = \mathbb{E}[\alpha_0 + \beta_0 X + A + V] = \alpha_0 + \beta_0 \cdot \frac{\beta_0 - \mu_C}{k_0} + \mu_A + 0$$

Step 3: Covariance Interlude.

Covariance of Linear Combinations

$$\text{Cov}(aX+bY,\ cW+dZ) = ac\ \text{Cov}(X,W)+ad\ \text{Cov}(X,Z)+bc\ \text{Cov}(Y,W)+bd\ \text{Cov}(Y,Z)$$

Step 4: Compute $\text{Cov}(A, X)$.

$$\text{Cov}(A, X) = \text{Cov}\left(A,\ \frac{1}{k_0}(\beta_0 - C)\right) = -\frac{1}{k_0}\text{Cov}(A, C) = -\frac{\sigma_{AC}}{k_0}$$

(The $\text{Cov}(A, \beta_0/k_0)$ term is zero since β_0/k_0 is a constant.)

Step 5: Compute $\text{Cov}(\log Y, X)$.

$$\text{Cov}(\log Y, X) = \text{Cov}(\beta_0 X + A + V,\ X) = \beta_0 \text{Var}(X) + \text{Cov}(A, X) + \underbrace{\text{Cov}(V, X)}_{=0}$$

$$= \beta_0 \cdot \frac{\sigma_C^2}{k_0^2} + \left(-\frac{\sigma_{AC}}{k_0}\right) = \frac{\beta_0 \sigma_C^2}{k_0^2} - \frac{\sigma_{AC}}{k_0}$$

Step 6: Compute $b_0 = \text{Cov}(\log Y, X)/\text{Var}(X)$.

$$b_0 = \frac{\dfrac{\beta_0 \sigma_C^2}{k_0^2} - \dfrac{\sigma_{AC}}{k_0}}{\dfrac{\sigma_C^2}{k_0^2}} = \beta_0 - k_0 \cdot \frac{\sigma_{AC}}{\sigma_C^2}$$

29.2 The Auxiliary Regression of A on X

The auxiliary regression (AR) coefficient $\pi_1 = \text{Cov}(A, X)/\text{Var}(X)$ is:

$$\pi_1 = \frac{\text{Cov}(A, X)}{\text{Var}(X)} = \frac{-\sigma_{AC}/k_0}{\sigma_C^2/k_0^2} = -k_0 \cdot \frac{\sigma_{AC}}{\sigma_C^2}$$

So the short regression is:

$$E^*[\log Y \mid X] = (a_0) + b_0 X, \qquad b_0 = \beta_0 + \pi_1$$

where $\pi_1 = -k_0 \sigma_{AC}/\sigma_C^2$.

Short Regression Coefficient in Card–Krueger

$$b_0 = \beta_0 - k_0 \frac{\sigma_{AC}}{\sigma_C^2}$$

With the full expressions for all terms:

$$a_0 = \alpha_0 + \mu_A + \frac{\sigma_{AC}}{\sigma_C^2}(\beta_0 - \mu_C)$$
$$b_0 = \beta_0 - k_0 \frac{\sigma_{AC}}{\sigma_C^2}$$

29.3 Ability Bias: When Does $b_0 \neq \beta_0$?

The direction of bias depends entirely on the sign of σ_{AC} (the covariance between ability A and cost C):

Ability Bias

$$b_0 > \beta_0 \iff \sigma_{AC} < 0 \quad (C_H \Rightarrow A_L : \text{high cost agents have low ability})$$
$$b_0 < \beta_0 \iff \sigma_{AC} > 0 \quad (C_H \Rightarrow A_H : \text{high cost agents have high ability})$$

Economic interpretation:

- **Case $\sigma_{AC} < 0$ ("ability bias" scenario):** Agents with high ability tend to face low costs of schooling (perhaps because schooling is easier for them, or they have access to financial aid). These high-ability agents optimally choose more schooling. An econometrician who regresses log earnings on schooling attributes some of the high-ability premium to schooling, *overstating* the true return β_0. This is the classic **upward ability bias**: $b_0 > \beta_0$.
- **Case $\sigma_{AC} > 0$:** High-ability agents face high costs (e.g., high opportunity cost of their time). They may choose less schooling despite high ability, causing the regression to *understate* β_0.

Graphical intuition: In the $(x, \log Y)$ space, the short regression line b_0 is steeper than the true slope β_0 when $\sigma_{AC} < 0$. This is because agents with more schooling also tend to have higher ability, so the observed earnings-schooling gradient conflates the effect of schooling *and* the effect of ability.

30 The Full Short Regression: Derivation via Long Run

30.1 Computing the Short Regression from the Long Run

Using the OVB formula with $W = A$ (ability) and $\pi_1 = \mathrm{Cov}(A, X)/\mathrm{Var}(X)$:

$$E^*[\log Y \mid X] = \alpha_0 + \beta_0 X + \underbrace{E^*[A \mid X]}_{\text{AR: } \pi_0 + \pi_1 X}$$

The auxiliary regression of A on X gives:

$$E^*[A \mid X] = \pi_0 + \pi_1 X, \quad \pi_1 = \frac{\mathrm{Cov}(A, X)}{\mathrm{Var}(X)}, \quad \pi_0 = \mu_A - \mathbb{E}[X]\pi_1$$

Substituting:

$$\begin{aligned} E^*[\log Y \mid X] &= \alpha_0 + \beta_0 X + \pi_0 + \pi_1 X \\ &= (\alpha_0 + \pi_0) + (\beta_0 + \pi_1) X \\ &= a_0 + b_0 X \end{aligned}$$

where:

$$a_0 = \alpha_0 + \pi_0 = \alpha_0 + \mu_A + \frac{\sigma_{AC}}{\sigma_C^2}(\beta_0 - \mu_C), \qquad b_0 = \beta_0 + \pi_1 = \beta_0 - k_0 \frac{\sigma_{AC}}{\sigma_C^2}$$

This is consistent with the direct computation in Section 6.

30.2 Structural Model → Data: The Full Chain

The Card–Krueger model shows how the economic structure generates the observed data distribution. The chain is:

$$(A, C, V^*) \xrightarrow{\text{Utility max}} X^* \xrightarrow{\text{Response fn}} \log Y(X^*) \xrightarrow{\text{observed}} F_{X,Y}$$

1. Distributions $F_{A,C}$ and F_{V^*} capture *all* human diversity that matters for the question.
2. The utility function, behavioral assumption (rational agents maximize expected utility), and information set generate the optimal choice X^*.
3. The potential earnings response function maps (X^*, A, V^*) to observed earnings Y.
4. The observed joint distribution $F_{X,Y}$ is the data we see.

This structural approach is powerful because it tells us *exactly* what the regression coefficient means in terms of the underlying economic parameters, and when it will or will not recover the true return to schooling β_0.

31 Summary: Key Formulas and Relationships

Object	Formula	Interpretation
BLP slope	$\beta_0 = \mathrm{Cov}(X, Y)/\mathrm{Var}(X)$	MSE-minimizing linear predictor slope
BLP intercept	$\alpha_0 = \mathbb{E}[Y] - \mathbb{E}[X]\beta_0$	Mean-preserving intercept
FOC 1	$\mathbb{E}[U] = 0$	Avg. prediction error is zero
FOC 2	$\mathbb{E}[UX] = 0 \iff \mathrm{Cov}(U, X) = 0$	Residual uncorrelated with X
OVB formula	$b_0 = \beta_0 + \gamma_0\pi_0$	Short = Long + (effect of W on Y) $\times$ (effect of X on W)
Frisch–Waugh	$E^*[Y \mid V] = \mathrm{const} + \gamma_0 V$	Coefficient on V = long-run γ_0
Optimal schooling	$X^* = (\beta_0 - C)/k_0$	MB = MC for schooling
Ability bias	$b_0 = \beta_0 - k_0\,\sigma_{AC}/\sigma_C^2$	Upward if $\sigma_{AC} < 0$

Big Picture

Linear regression (BLP) is defined purely as the MSE-minimizing linear predictor—no causality assumptions needed. The OVB formula tells us how and when the short regression coefficient b_0 differs from the long regression coefficient β_0. The Card–Krueger structural model embeds the regression in an economic model, showing that the short regression conflates the true return β_0 with the ability-schooling correlation, generating "ability bias." The direction and magnitude of this bias depends on σ_{AC}, the covariance between ability and cost.

Multivariate BLP, Bayesian Bootstrap, and Proxy Variables

32 Linear Regression: The General (Multivariate) Case

32.1 Setup

We now extend linear regression from the bivariate case (Y on a single X) to the fully general multivariate case with k regressors. Let:

- $(X, Y) \sim F_{X,Y}$, where $X = (X_1, \ldots, X_k)'$ is a $k \times 1$ vector of regressors and Y is a scalar outcome.
- Example: $X = (\text{age}, \text{education}, \text{AFQT})' \in \mathbb{R}^k$, $Y = \log(\text{earnings})$.
- We augment X with a constant by defining the **regressor vector**:

$$R = \begin{pmatrix} 1 \\ X \end{pmatrix}, \quad (k+1) \times 1$$

  and the **coefficient vector**:

$$\eta = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad (k+1) \times 1$$

  so the linear predictor is $R'\eta = \alpha + X'\beta$.

32.2 The Multivariate MSE Problem

The Best Linear Predictor (BLP) of Y given X (equivalently, given R) minimizes:

$$\eta_0 = \arg\min_{\eta \in \mathbb{R}^{k+1}} \mathbb{E}\big[(Y - R'\eta)^2\big]$$

This is the MSE-minimizing linear predictor of Y given $(1, X)$.

32.3 First-Order Conditions (FOCs) and the Normal Equations

Differentiating the objective with respect to η and setting to zero yields the **population normal equations**:

$$\mathbb{E}[RR']\,\eta - \mathbb{E}[RY] = 0 \quad \Longrightarrow \quad \mathbb{E}[RR']\,\eta = \mathbb{E}[RY]$$

Dimensions:

- $\mathbb{E}[RR']$: $(k+1) \times (k+1)$ matrix.
- $\mathbb{E}[RY]$: $(k+1) \times 1$ vector.
- This is a system of $k+1$ linear equations in $k+1$ unknowns.

Premultiplying both sides by $\mathbb{E}[RR']^{-1}$ (assuming invertibility):

Multivariate BLP Solution

$$\eta_0 = \mathbb{E}[RR']^{-1}\,\mathbb{E}[RY]$$

The Best Linear Predictor of Y given $(1, X)$ is:

$$E^*[Y \mid 1, X] = \alpha_0 + X'\beta_0 = R'\eta_0$$

32.4 Solving via the Partitioned Inverse (Henry Theil)

We need to compute η_0 explicitly. Using the notation $\frac{1}{A} = A^{-1}$, we expand:

$$\mathbb{E}[RR'] = \mathbb{E}\left[\begin{pmatrix}1\\X\end{pmatrix}(1, X')\right] = \mathbb{E}\left[\begin{pmatrix}1 & X'\\X & XX'\end{pmatrix}\right] = \begin{pmatrix}1 & \mathbb{E}[X]'\\\mathbb{E}[X] & \mathbb{E}[XX']\end{pmatrix}$$

This is a partitioned matrix with: $A = 1$, $B = \mathbb{E}[X]'$, $C = \mathbb{E}[X]$, $D = \mathbb{E}[XX']$.
By the **Partition Inverse Formula**:

$$\begin{pmatrix}A & B\\C & D\end{pmatrix}^{-1} = \begin{pmatrix}A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1} & -A^{-1}B(D - CA^{-1}B)^{-1}\\-(D - CA^{-1}B)^{-1}CA^{-1} & (D - CA^{-1}B)^{-1}\end{pmatrix}$$

The key quantity is the **Schur complement**:

$$D - CA^{-1}B = \mathbb{E}[XX'] - \mathbb{E}[X]\mathbb{E}[X]' \;=\; \mathbb{V}(X) \;\equiv\; \mathbb{V}(x)$$

which is the $k \times k$ **variance–covariance matrix** of X.

Expanded variance–covariance matrix:

$$\mathbb{V}(X) = \begin{pmatrix}\mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \cdots & \mathrm{Cov}(X_1, X_k)\\\mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & & \vdots\\\vdots & & \ddots & \vdots\\\mathrm{Cov}(X_k, X_1) & \cdots & \cdots & \mathrm{Var}(X_k)\end{pmatrix}$$

Similarly, define the **covariance vector** of X with Y:

$$C(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y], \quad k \times 1$$

Working through the full partition inverse and the expression for $\mathbb{E}[RY] = \big(\mathbb{E}[Y],\ \mathbb{E}[XY]\big)'$:

Explicit Multivariate BLP Coefficients

$$\beta_0 = \mathbb{V}(X)^{-1}\,C(X, Y) \qquad \text{(Covariance / Variance matrix)}$$

$$\alpha_0 = \mathbb{E}[Y] - \mathbb{E}[X]'\beta_0$$

The BLP is:

$$E^*[Y \mid 1, X] = \alpha_0 + X'\beta_0$$

Remark 32.1. This is the exact multivariate generalization of the bivariate formulas $\beta_0 = \mathrm{Cov}(X, Y)/\,\mathrm{Var}(X)$ and $\alpha_0 = \mathbb{E}[Y] - \mathbb{E}[X]\beta_0$. The scalar $\mathrm{Var}(X)$ becomes the matrix $\mathbb{V}(X)$ and the scalar $\mathrm{Cov}(X, Y)$ becomes the vector $C(X, Y)$.

32.5 Orthogonality Conditions and Residual

The prediction error is $U = Y - \alpha_0 - X'\beta_0$. The FOCs are equivalent to:

$$\mathbb{E}[U] = 0, \qquad \mathbb{E}[XU] = C(X, U) = 0 \quad (k \times 1)$$

Together: $\mathbb{E}[RU] = 0$, i.e., the residual is uncorrelated with every element of $R = (1, X')'$.

An equivalent demeaned expression for the residual:

$$U = (Y - \mathbb{E}[Y]) - (X - \mathbb{E}[X])'\beta_0$$

32.6 Sample Analogue (OLS)

Given data $\{(x_i, y_i)\}_{i=1}^n$, the OLS estimator is:

$$\hat{\beta} = \left[\frac{1}{n}\sum_{i=1}^n x_i x_i'\right]^{-1} \left[\frac{1}{n}\sum_{i=1}^n x_i y_i\right]$$

which replaces population moments with sample averages. This is the finite-sample estimate of β_0.

33 The Bayesian Bootstrap

33.1 Motivation: A Population of Populations

The **Bayesian Bootstrap** is a method for performing Bayesian inference about arbitrary functionals of a population distribution—in particular, for regression coefficients—without requiring a parametric likelihood. The key metaphor is a **pot of Roman coins**.

Metaphor: Imagine a large pot containing Roman coins, each with a different probability θ of landing heads when flipped. A coin is drawn at random from the pot and flipped N times. The goal is to learn about θ (the head probability of the selected coin) from the observed flip outcomes.

Formally:

- The pot represents a "population of populations" or a prior over data-generating processes.
- $\theta \sim \Pi(\theta)$: the distribution of head probabilities among coins in the pot (*prior*).
- Each coin can be thought of as a "population" (a distribution, sampling mechanism, etc.) that produces heads with probability θ and tails with probability $1 - \theta$.
- Real application: θ = fraction of neighborhoods voting Democrat vs. Republican.

33.2 The Beta–Binomial Model (Coin Flip Version)

Step 1: Pick a coin and flip it.

1. Pick a coin from the pot with head probability θ.
2. Flip it N times, recording outcomes $Z = (Z_1, \ldots, Z_N)'$, where $Z_i \in \{0, 1\}$.
3. $S_N = \sum_{i=1}^{N} Z_i$ = number of successes (heads).

Likelihood:

$$\ell(Z \mid \theta) = \prod_{i=1}^{N} \theta^{Z_i}(1-\theta)^{1-Z_i} = \theta^{S_N}(1-\theta)^{N-S_N}$$

Prior:

$$\theta \sim \Pi(\theta) = \text{Beta}(\alpha_1, \alpha_2)$$

The Beta distribution describes the fraction of coins in the hoard with head probability between θ and $\bar{\theta}$. Different prior choices:

- **Laplace prior:** $\alpha_1 = \alpha_2 = 1 \Rightarrow$ uniform $U(0, 1)$.
- **Improper (uninformative) prior:** $\alpha_1 = \alpha_2 = 0$ ("Haldane Prior"). Technically not a proper distribution; we work with the limiting posterior as $\alpha_1, \alpha_2 \to 0$.

Posterior (via Bayes' Rule):

$$\Pi(\theta \mid Z = z) = \frac{\ell(z \mid \theta)\,\Pi(\theta)}{\int_0^1 \ell(z \mid t)\,\Pi(t)\,dt} = \frac{\theta^{S_N+\alpha_1-1}(1-\theta)^{N-S_N+\alpha_2-1}}{B(S_N+\alpha_1,\ N-S_N+\alpha_2)}$$

Beta–Binomial Posterior

$$\theta \mid Z \ \sim\ \text{Beta}(S_N+\alpha_1,\ \ N-S_N+\alpha_2)$$

Interpretation: The prior parameters α_1, α_2 are pseudo-counts of "hypothetical" previously observed heads (successes) and tails (failures). After seeing S_N successes and $N - S_N$ failures in real data, we simply add them to the prior counts.

33.3 Posterior Simulation via Gamma Weights

Computing the posterior analytically is one option; the Bayesian Bootstrap provides a **simulation-based** alternative that generalizes to arbitrary estimands (including regression coefficients).

Key fact (Gamma–Beta relationship): If $X \sim \text{Gamma}(\alpha, b)$ and $Y \sim \text{Gamma}(\beta, b)$ independently, then:

$$\frac{X}{X+Y} \sim \text{Beta}(\alpha, \beta)$$

Algorithm: Posterior simulation for θ. For each simulation draw $s = 1, \ldots, S$:

1. Draw N i.i.d. exponentials (Gamma(1,1) draws):

$$W_i^{(s)} \overset{\text{i.i.d.}}{\sim} \text{Gamma}(1,1), \quad i = 1, \ldots, N$$

2. By the **reproductive (stable) property** of the Gamma distribution:

$$\sum_{i=1}^{N} W_i^{(s)} Z_i \sim \text{Gamma}(S_N, 1), \qquad \sum_{i=1}^{N} W_i^{(s)}(1 - Z_i) \sim \text{Gamma}(N - S_N, 1)$$

3. Define normalized weights:

$$V_i^{(s)} = \frac{W_i^{(s)}}{\sum_{j=1}^{N} W_j^{(s)}}$$

   Note $\sum_i V_i^{(s)} = 1$ and $\mathbb{E}[V_i] = 1/N$.

4. The simulation draw of θ is:

$$\tilde{\theta}^{(s)} = \sum_{i=1}^{N} V_i^{(s)} Z_i \;\sim\; \text{Beta}(S_N, N - S_N) \quad \text{(posterior draw)}$$

Key Comparison: MLE vs. Bayesian Bootstrap

$$\hat{\theta}_{\text{MLE}} = \frac{1}{N} \sum_{i=1}^{N} Z_i \qquad \text{(equal weights } 1/N\text{)}$$

$$\tilde{\theta}^{(s)} = \sum_{i=1}^{N} V_i^{(s)} Z_i \qquad \text{(random Gamma weights)}$$

Both are weighted averages of the data. The MLE uses the "empirical measure" (flat weights); the Bayesian Bootstrap uses a "simulated measure" (random Dirichlet weights from Gamma draws). When $S \to \infty$, the distribution of $\tilde{\theta}^{(s)}$ converges to the true posterior Beta$(S_N, N - S_N)$.

33.4 Posterior Summaries via Simulation

From S draws $\tilde{\theta}^{(1)}, \ldots, \tilde{\theta}^{(S)}$:

Posterior mean (numerical integration):

$$\hat{\mathbb{E}}[\theta \mid Z] = \frac{1}{S} \sum_{s=1}^{S} \tilde{\theta}^{(s)} \approx \mathbb{E}[\theta \mid Z] = \frac{S_N}{N}$$

Posterior variance:

$$\hat{\mathbb{V}}[\theta \mid Z] = \frac{1}{S}\sum_{s=1}^{S}\left(\tilde{\theta}^{(s)} - \hat{\mathbb{E}}[\theta \mid Z]\right)^2$$

Posterior credible interval (Bayesian confidence interval): For $\alpha = 0.05$, the 95% **posterior credible interval** $[\underline{\theta}, \bar{\theta}]$ satisfies:

$$\Pr(\theta \in [\underline{\theta}, \bar{\theta}] \mid Z) = 1 - \alpha = 0.95$$

Interpretation: "I believe there is a 95% chance that θ is between $\underline{\theta}$ and $\bar{\theta}$." This is a direct probability statement about the parameter, unlike a frequentist confidence interval.

Posterior expectation of any function $g(\theta)$:

$$\hat{\mathbb{E}}[g(\theta) \mid Z] = \frac{1}{S}\sum_{s=1}^{S} g(\tilde{\theta}^{(s)})$$

For example, $g(\theta) = e^{\theta} - \pi\theta + \theta^{\pi} - \sqrt{\theta}$ is computed easily by just applying g to each simulation draw.

33.5 Bayesian Bootstrap for Regression Coefficients

The Bayesian Bootstrap extends immediately to the multivariate regression setting. Now the data are $Z_i = (X_i', Y_i)'$, the full observation for individual i, forming the dataset $\mathcal{Z} = \{z_1, \ldots, z_N\}$.

The population regression coefficient:

$$\beta_0 = \mathbb{E}[XX']^{-1}\mathbb{E}[XY] = \left[\sum_{j=1}^{J}\theta_j x_j x_j'\right]^{-1}\left[\sum_{j=1}^{J}\theta_j x_j y_j\right]$$

where $\theta_j = \Pr(Z = z_j \mid \underline{\theta})$ is the population frequency of type-j individuals.

The OLS estimator is:

$$\hat{\beta} = \left[\frac{1}{N}\sum_{i=1}^{N} x_i x_i'\right]^{-1}\left[\frac{1}{N}\sum_{i=1}^{N} x_i y_i\right]$$

Bayesian Bootstrap for Regression (Algorithm)

For $s = 1, \ldots, S$:

1. Draw N i.i.d. Gamma(1,1) weights: $W_i^{(s)} \overset{\text{i.i.d.}}{\sim} \text{Gamma}(1,1)$, $i = 1, \ldots, N$.
2. Normalize: $V_i^{(s)} = W_i^{(s)} / \sum_{j=1}^{N} W_j^{(s)}$.

3. Compute the weighted regression coefficient:

$$\tilde{\beta}^{(s)} = \left[\sum_{i=1}^{N} V_i^{(s)} x_i x_i'\right]^{-1} \left[\sum_{i=1}^{N} V_i^{(s)} x_i y_i\right]$$

This is a **Weighted Least Squares (WLS)** regression of Y_i on X_i with weights $V_i^{(s)}$.

Use $\tilde{\beta}^{(1)}, \ldots, \tilde{\beta}^{(S)}$ to summarize any feature of the posterior $\Pi(\beta \mid \mathcal{Z})$.

Key properties:

- $\tilde{\beta}^{(s)} \sim \Pi(\beta \mid \mathcal{Z})$: each draw is approximately a draw from the posterior of β.
- $\hat{\mathbb{E}}[\beta \mid \mathcal{Z}] = \frac{1}{S}\sum_s \tilde{\beta}^{(s)} \approx \hat{\beta}$ (OLS): the posterior mean of β is approximately the OLS estimate.
- The bootstrap is a way of **reweighting** the data. The "smooth bootstrap" (using Gamma weights) allows you to be fully Bayesian about the regression coefficients without specifying a parametric likelihood.

34 Proxy Variable Regression

34.1 The Problem: Unobserved Ability

In the schooling–earnings context, the causal long regression of interest includes unobserved ability A:

Definition 34.1 (Infeasible Oracle Regression (LR#))**.**

$$Y = \alpha_0 + \beta_0 X + \gamma_0 A + U \qquad \text{(LR\#)}$$

where:

- Y = log earnings
- X = years of schooling (policy / treatment variable)
- A = ability to do well in the labor market (latent; "keeps us up at night")
- U = luck and other unobservables, orthogonal to everything

U is partitioned into: (a) stuff we worry about (correlated with X, A) and (b) stuff that drives change but is orthogonal to our variables (we don't worry about this part).

The assumption that LR# is **structural/causal** means β_0 is a policy-relevant parameter: it measures the effect of changing X on Y, holding A fixed. This requires looking at small changes in X that don't induce general equilibrium effects.

The problem: A is **unobserved**. If we simply regress Y on X (short regression), we get $b_0 \neq \beta_0$ due to omitted variable bias.

34.2 Proxy Variables

Instead of A directly, we observe W, a vector of **proxy variables** for ability:

$$W = (\text{AFQT score}, \text{ KWW score}, \text{ Big Five personality}, \text{ parental schooling})$$

Definition 34.2 (Feasible "Kitchen-Sink" Regression (PVR))**.**

$$E^*[Y \mid X, W] = a_0 + b_0 X + c_0 W \qquad \text{(PVR / Proxy Variable Regression)}$$

The central question: **under what assumptions does** $b_0 = \beta_0$?

34.3 Two Key Assumptions

Setup: Start from:

$$E^*[Y \mid X, W] = E^*[\alpha_0 + \beta_0 X + \gamma_0 A + U \mid X, W] = \alpha_0 + \beta_0 X + \gamma_0 E^*[A \mid X, W] + E^*[U \mid X, W]$$

To make progress, two assumptions are imposed:

Definition 34.3 (Assumption A.1: Redundancy)**.**

$$E^*[Y \mid X, A, W] = E^*[Y \mid X, W]$$

Interpretation: If we already know A, knowing W provides no additional predictive value for Y. Equivalently, $E^*[U \mid X, W, A] = 0$ (the error is unpredictable from any of X, W, or A).

This means W is a sufficient statistic for A in predicting Y (given X).

Definition 34.4 (Assumption A.2: Conditional Unconnectedness)**.**

$$E^*[A \mid X, W] = E^*[A \mid W]$$

Also called: **unconfoundedness**, **exogeneity**, **selection on observables**, or **conditional independence assumption**.

Interpretation: Conditional on the proxy variables W, X does not additionally help predict A. In other words, once we control for W, the policy variable X is uncorrelated with ability A.

Define the residual $V = X - E^*[X \mid W]$ (variation in X unpredictable from W). Then A.2 implies $\text{Cov}(A, V) = 0$.

When is A.2 plausible?

X is the policy variable (schooling); A is ability (a worry, as agents choose X endogenously based on ability). If X and A are correlated due to the agent's choice, we have OVB. A.2 is more plausible when W is closely related to the agent's **information set** at the time they choose X: if W captures everything about ability that influenced the

schooling decision, then X is "as good as randomly assigned" conditional on W.

34.4 Main Result: $b_0 = \beta_0$ under A.1 and A.2

Let the auxiliary regression of A on W be:

$$E^*[A \mid W] = \lambda_0 + \eta_0 W \qquad \text{(A.R.)}$$

Under A.2: $E^*[A \mid X, W] = E^*[A \mid W] = \lambda_0 + \eta_0 W$.
Substituting into the PVR and applying the Law of Iterated Linear Predictors (LILP):

$$\begin{aligned}
E^*[Y \mid X, W] &= \alpha_0 + \beta_0 X + \gamma_0\, E^*[A \mid X, W] + E^*[E^*[U \mid X, W, A] \mid X, W] \\
&\underset{A.1}{=} \alpha_0 + \beta_0 X + \gamma_0\, E^*[A \mid X, W] + 0 \\
&\underset{A.2}{=} \alpha_0 + \beta_0 X + \gamma_0\, E^*[A \mid W] \\
&= \alpha_0 + \beta_0 X + \gamma_0(\lambda_0 + \eta_0 W) \\
&= \underbrace{(\alpha_0 + \gamma_0\lambda_0)}_{a_0} + \underbrace{\beta_0}_{b_0} X + \underbrace{\gamma_0\eta_0}_{c_0} W
\end{aligned}$$

Proxy Variable Regression Theorem

Under Assumptions A.1 (Redundancy) and A.2 (Conditional Unconnectedness):

$$b_0 = \beta_0$$

The coefficient on X in the feasible proxy variable regression equals the causal coefficient in the unobservable long regression. There is **no OVB** in the PVR coefficient on X.

34.5 Verification in the Card–Krueger Model

Recall the Card–Krueger setup:

- Earnings function: $Y(x) = \alpha_0 + \beta_0 X + \gamma_0 A + V$
- Cost function: $Z(x) = Cx + \frac{k_0}{2}x^2$
- Optimal schooling: $X = \frac{1}{k_0}(\beta_0 - C)$

A.1 (Redundancy): $E^*[V \mid A, C, W] = E^*[V \mid A, C] = 0$ since $V \perp (A, C)$ (luck is independent of ability and cost). W is redundant given (A, C) for predicting V.

A.2 (Conditional Unconnectedness): We need $E^*[A \mid C, W] = E^*[A \mid W]$.

Since $X = \frac{1}{k_0}(\beta_0 - C)$, X is a linear function of C (and vice versa). So $E^*[A \mid X, W] = E^*[A \mid C, W]$. The key is the **timing assumption**:

1. Agent observes C and W.
2. Agent chooses $X = \frac{1}{k_0}(\beta_0 - C)$.
3. Y is realized.

If W captures everything about C that predicts A — i.e., the residual $\tilde{Z} = C - E^*[C \mid W]$ has $\mathrm{Cov}(\tilde{Z}, A) = 0$ — then conditional on W, X (or equivalently C) does not additionally predict A. This is achieved when $(A, C) \mid W$ has a block-diagonal covariance:

$$\begin{pmatrix} A \\ C \end{pmatrix} \Bigg| W \sim N\left(\begin{pmatrix} \mu_A(W) \\ \mu_C(W) \end{pmatrix}, \begin{pmatrix} \sigma_A^2(W) & 0 \\ 0 & \sigma_C^2(W) \end{pmatrix} \right)$$

i.e., $A \perp\!\!\!\perp C \mid W$ (ability and cost are conditionally independent given the proxies). Under this assumption, $E^*[A \mid C, W] = E^*[A \mid W]$, confirming A.2.

35 The Bias–Variance Tradeoff

35.1 Prediction Risk and IMSE

Let $\hat{m}(x)$ be an estimated regression function (e.g., OLS, kernel regression, a neural network). Let Y^* be a new draw of Y at the observed design points $\{x_1, \ldots, x_N\}$, holding design fixed.

Definition 35.1 (Prediction Risk and IMSE).

- **Prediction risk:** $\mathbb{E}[\|Y^* - \hat{m}\|^2]$ — expected squared distance between new outcomes and fitted values.
- **IMSE (Integrated Mean Squared Error):** $\mathbb{E}[\|\hat{m} - m\|^2]$ — expected squared distance between the estimated and true regression function.

These are related by:

$$\mathbb{E}[\|Y^* - \hat{m}\|^2] \doteq N\sigma^2 + \mathbb{E}[\|\hat{m} - m\|^2]$$

where $\sigma^2 = \mathrm{Var}(Y \mid X)$ is the irreducible noise variance. The two are **isomorphic**: minimizing prediction risk is equivalent to minimizing IMSE (the $N\sigma^2$ term is fixed).

35.2 Bias–Variance Decomposition

The IMSE decomposes into two terms:

Bias–Variance Tradeoff

$$\mathbb{E}[\|\hat{m} - m\|^2] = \underbrace{\sum_{i=1}^{N} \mathrm{Var}(\hat{m}(x_i))}_{\text{Variance}} + \underbrace{\sum_{i=1}^{N} \left(\mathbb{E}[\hat{m}(x_i)] - m(x_i)\right)^2}_{\text{Bias}^2}$$

- **Variance:** How much does $\hat{m}(x_i)$ fluctuate across different datasets? High variance means the estimator is sensitive to which sample was drawn (overfitting).
- **Bias^2:** How far is the expected estimate $\mathbb{E}[\hat{m}(x_i)]$ from the true $m(x_i)$? High bias means the model is systematically wrong (underfitting).

The tradeoff:

- More complex models (more parameters, less regularization) have **lower bias but higher variance**: they can fit the true function more flexibly but overfit to noise.
- Simpler models (e.g., restricting to linear functions) have **higher bias but lower variance**: they may miss nonlinearities but are more stable across samples.
- The optimal model minimizes the sum Variance + Bias^2.

Example 35.2 (OLS vs. Saturated Model)**.** OLS with k regressors imposes a linear functional form. It has low variance (only $k+1$ parameters to estimate) but potentially high bias if the true $m(x)$ is nonlinear. A saturated (fully nonparametric) model has zero bias but high variance (it overfits every sample). The optimal choice of complexity depends on the sample size N relative to the complexity of the true function.

36 Summary Table

Object	Formula	Interpretation
Multivariate BLP	$E^*[Y\|1, X] = \alpha_0 + X'\beta_0$	MSE-minimizing linear predictor
β_0	$\mathbb{V}(X)^{-1}C(X, Y)$	Cov/Var matrix formula
α_0	$\mathbb{E}[Y] - \mathbb{E}[X]'\beta_0$	Mean-preserving intercept
Normal equations	$\mathbb{E}[RR']\eta_0 = \mathbb{E}[RY]$	Population FOCs
BB weights	$V_i^{(s)} = W_i^{(s)}/\sum_j W_j^{(s)}$, $W_i \sim$ Gamma$(1, 1)$	Random reweighting of data
BB estimate	$\tilde{\beta}^{(s)} = (\sum V_i^{(s)} x_i x_i')^{-1}(\sum V_i^{(s)} x_i y_i)$	WLS draw from posterior
A.1 Redundancy	$E^*[Y\|X, A, W] = E^*[Y\|X, W]$	W subsumes A for predicting Y
A.2 Cond. Unconn.	$E^*[A\|X, W] = E^*[A\|W]$	$X \perp A \mid W$
PVR result	$b_0 = \beta_0$ under A.1 & A.2	Proxy regression recovers causal coeff.
Bias–Variance	$\mathbb{E}[\|\hat{m} - m\|^2] = \text{Var} + \text{Bias}^2$	Decomposition of IMSE

Big Picture

Three interconnected ideas: (1) The multivariate BLP generalizes the bivariate case to $\beta_0 = \mathbb{V}(X)^{-1}C(X, Y)$, a matrix-vector analog of covariance over variance. (2) The Bayesian Bootstrap provides a simulation-based way to characterize the entire posterior distribution of β (or any other estimand) by reweighting observations with Gamma random variables — bridging the gap between Bayesian and frequentist inference without specifying a parametric likelihood. (3) Proxy variable regression shows that if proxy variables W satisfy Redundancy (A.1) and Conditional Unconnectedness (A.2), the feasible PVR coefficient b_0 recovers the causal coefficient β_0 from the infeasible oracle regression — providing a rigorous justification for "kitchen-sink" regressions with many controls.

Model Selection, IMSE, SURE, Kernel Regression, and Trees

37 Setup: The Nonparametric Regression Problem

37.1 Framework

We observe N data points (x_i, Y_i) where the outcomes follow:

$$Y_i = m(x_i) + u_i, \quad u_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2), \quad i = 1, \ldots, N$$

- $Y_i \in \mathcal{Y} \subset \mathbb{R}$: scalar outcome (e.g., end-of-first-year college GPA).
- $x_i \in \mathcal{X} \subset \mathbb{R}^k$: regressor vector (e.g., HS GPA, SAT score, class rank).
- $m : \mathcal{X} \to \mathcal{Y}$: unknown regression function.
- The design matrix $\mathcal{X} = (x_1, x_2, \ldots, x_N)'$ is treated as **non-stochastic** (fixed in design across repeated samples). This simplifies the analysis: we average over Y only, holding x_i fixed.

Goal: Construct an estimate $\hat{m}(x_i)$ of the true $m(x_i)$ at each observed design point. Collect these into a vector:

$$m = \begin{pmatrix} m(x_1) \\ \vdots \\ m(x_N) \end{pmatrix}_{N\times 1}, \qquad \hat{m} = \begin{pmatrix} \hat{m}(x_1) \\ \vdots \\ \hat{m}(x_N) \end{pmatrix}_{N\times 1}$$

Special estimators:

- **Constant estimator:** $\hat{m}(x_i) = \bar{Y} = \frac{1}{N}\sum_i Y_i$ for all i ("everyone gets a 3.0"). High bias, zero variance.
- **OLS:** $\hat{m}(x_i) = x_i'\hat{\beta}$, where $\hat{\beta} = \left(\sum_i x_i x_i'\right)^{-1} \sum_i x_i Y_i$.
- **Correct the dots (CTD):** $\hat{m}(x_i) = Y_i$ for all i (interpolate). Zero in-sample error, but entirely noise-driven.

The Euclidean norm of a vector m is $\|m\| = \left(\sum_{i=1}^N m_i^2\right)^{1/2}$ and measures the spread of the fitted values.

38 Apparent Error and the IMSE Decomposition

38.1 Apparent Error (Efron, 2004)

Definition 38.1 (Apparent Error)**.** The **apparent error** (also called in-sample fit or empirical risk) is:

$$\mathbb{E}\left[\|Y - \hat{m}\|^2\right] = \mathbb{E}\left[\sum_{i=1}^N (Y_i - \hat{m}(x_i))^2\right]$$

This is the expected squared distance between outcomes and fitted values, averaged over the randomness in Y (with design fixed).

Apparent error is **low** when we use procedures that closely fit the sample (e.g., CTD has apparent error $= 0$ technically). But low apparent error is not the same as good out-of-sample performance.

38.2 Decomposing Apparent Error

We write:

$$Y - \hat{m} = (Y - m) - (\hat{m} - m)$$

and expand the squared norm:

$$\begin{aligned}\mathbb{E}\big[\|Y - \hat{m}\|^2\big] &= \mathbb{E}\big[\|(Y - m) - (\hat{m} - m)\|^2\big] \\ &= \underbrace{\mathbb{E}\big[\|Y - m\|^2\big]}_{1} + \underbrace{\mathbb{E}\big[\|\hat{m} - m\|^2\big]}_{2} - 2\underbrace{\mathbb{E}\big[(Y - m)'(\hat{m} - m)\big]}_{3}\end{aligned}$$

Term 1: Noise floor.

$$\mathbb{E}\big[\|Y - m\|^2\big] = \mathbb{E}\left[\sum_i (Y_i - m(x_i))^2\right] = \sum_i \mathbb{E}\big[u_i^2\big] = \sum_i \sigma^2 = N\sigma^2$$

This is irreducible noise — no estimator can do better than this floor.

Term 2: IMSE.

$$\mathbb{E}\big[\|\hat{m} - m\|^2\big] = \sum_{i=1}^{N} \mathbb{E}\big[(\hat{m}(x_i) - m(x_i))^2\big] \;\equiv\; \mathrm{IMSE}(\hat{m}, m)$$

This is the **Integrated Mean Squared Error** (IMSE), a measure of the overall quality of the estimator. The expectation in IMSE averages over the randomness in Y (i.e., across all possible datasets with the same design $x_1, \ldots, x_N$).

Term 3: The cross term. Since $m = m(x_i)$ is non-random (design is fixed) and $\hat{m}$ typically depends on Y but $Y - m = u$ is the error:

$$\begin{aligned}\mathbb{E}\big[(Y - m)'(\hat{m} - m)\big] &= -2\mathbb{E}\big[(Y - m)'\hat{m}\big] + 2\mathbb{E}\big[(Y - m)'\big] \cdot m \\ &= -2\mathbb{E}\big[(Y - m)'\hat{m}\big] + 0\end{aligned}$$

(since $\mathbb{E}[Y - m] = \mathbb{E}[u] = 0$).

This simplifies to:

$$-2\mathbb{E}\big[(Y - m)'\hat{m}\big] = -2\sum_{i=1}^{N} \mathbb{E}\big[(Y_i - m(x_i))\hat{m}(x_i)\big] = -2\sum_{i=1}^{N} \mathrm{Cov}(Y_i, \hat{m}(x_i))$$

Definition 38.2 (Degrees of Freedom)**.** The **degrees of freedom** of an estimator $\hat{m}$ are defined as:

$$\mathrm{df}(\hat{m}) \stackrel{\text{def}}{=} \sum_{i=1}^{N} \frac{\mathrm{Cov}(Y_i, \hat{m}(x_i))}{\sigma^2}$$

Alternative names: *expected in-sample fit*, *model complexity.*

Intuition:

- If $\hat{m}$ is highly responsive to each Y_i (like CTD), then $\mathrm{Cov}(Y_i, \hat{m}(x_i))$ is large and $\mathrm{df}(\hat{m})$ is high.
- If $\hat{m}$ ignores the data (constant estimator), Cov $= 0$ and df $= 0$.
- Overfitting $\Rightarrow$ high degrees of freedom / model complexity.
- We want a balance between apparent error and degrees of freedom.

38.3 The Master IMSE Formula

Assembling all three terms:

$$\mathbb{E}\big[\|Y - \hat{m}\|^2\big] = N\sigma^2 + \mathrm{IMSE}(\hat{m}, m) - 2\sigma^2 \ \mathrm{df}(\hat{m})$$

Rearranging to isolate IMSE ("Risk"):

IMSE

$$\underbrace{\sum_{i=1}^{N} \mathbb{E}\big[(\hat{m}(x_i) - m(x_i))^2\big]}_{\text{IMSE / "Risk"}} = -N\sigma^2 + \underbrace{\sum_{i=1}^{N} \mathbb{E}\big[(Y_i - \hat{m}(x_i))^2\big]}_{\text{Apparent Error}} + 2\sigma^2 \ \mathrm{df}(\hat{m})$$

IMSE is the negative of expected utility (Risk $= -$Expected Utility).

This formula has a crucial interpretation: risk equals apparent error minus the noise floor, plus a penalty for model complexity. A model that perfectly fits the training data has zero apparent error but high degrees of freedom, and thus high IMSE.

38.4 Constant Estimator: A Sanity Check

For $\hat{m}(x_i) = \bar{Y} = \frac{1}{N}\sum_j Y_j$:

$$\sum_{i=1}^{N} \mathbb{E}\big[(\bar{Y} - m(x_i))^2\big] = \sum_{i=1}^{N} \left[\left(m(x_i) - \frac{1}{N}\sum_j m(x_j) \right)^2 + \sigma^2 \right]$$

The first term is the squared deviation of each $m(x_i)$ from the average true signal (bias^2) and the second is the variance contribution. This confirms the bias-variance decomposition.

39 Risk for Linear Estimators and Series Regression

39.1 Linear Estimators and the Hat Matrix

Definition 39.1 (Linear Estimator)**.** An estimator $\hat{m}$ is **linear** if:

$$\hat{m} = H \cdot Y, \quad H = [h_{ij}]_{1 \le i,j \le N}, \quad H \text{ is } N \times N \text{ and non-stochastic}$$

H is called the **hat matrix** (or smoothing matrix). It maps the outcome vector Y to fitted values.

For linear estimators, degrees of freedom have a particularly clean form:

Degrees of Freedom for Linear Estimators

$$\mathrm{df}(\hat{m}) = \sum_{i=1}^{N} h_{ii} = \mathrm{Tr}(H)$$

Proof: $\mathrm{Cov}(Y_i, \hat{m}(x_i)) = \mathrm{Cov}(Y_i, \sum_j h_{ij} Y_j) = \sum_j h_{ij} \mathrm{Cov}(Y_i, Y_j) = h_{ii}\sigma^2$ (since Y_i are independent). So $\mathrm{df}(\hat{m}) = \sum_i h_{ii}\sigma^2/\sigma^2 = \mathrm{Tr}(H)$.

The IMSE for a linear estimator $\hat{m} = HY$ is therefore:

$$\mathrm{IMSE}(\hat{m}, m) = -N\sigma^2 + \sum_{i=1}^{N} \mathbb{E}[(Y_i - \hat{m}(x_i))^2] + 2\sigma^2 \,\mathrm{Tr}(H)$$

39.2 Series (Basis Function) Regression

A flexible approach to nonparametric estimation is to project Y onto a J-dimensional **basis** of functions.

Definition 39.2 (Basis Functions)**.** A basis function vector $\lambda_J(x) : \mathbb{R}^k \to \mathbb{R}^J$ maps each design point to a $J \times 1$ vector of features. Two important examples:

- **Polynomial basis** (for $x_i \in [0, 1]$, scalar):

$$\lambda_J(x) = (1, x, x^2, \ldots, x^{J-1})'$$

- **Orthogonal cosine basis** (for $x_i \in [0, 1]$):

$$\lambda_J(x) = \left(1, \sqrt{2}\cos(\pi x), \sqrt{2}\cos(2\pi x), \ldots, \sqrt{2}\cos((J-1)\pi x)\right)'$$

  These satisfy $\int_0^1 \cos(\pi n x)\cos(\pi m x)\,dx = 0$ for $n \neq m$, making the basis orthonormal.

Define the $N \times J$ design matrix of basis functions:

$$W_J = \begin{pmatrix} \lambda_J(x_1)' \\ \vdots \\ \lambda_J(x_N)' \end{pmatrix}$$

The OLS fit of Y onto $\lambda_J(x_i)$ gives:

$$\hat{\pi}_J = (W_J'W_J)^{-1}W_J'Y \quad (J \times 1), \qquad \hat{m}_J = W_J\hat{\pi}_J = W_J(W_J'W_J)^{-1}W_J'Y = H_JY$$

where $H_J = W_J(W_J'W_J)^{-1}W_J'$ is the **projection matrix** (hat matrix).

Degrees of freedom for series regression:

$$\begin{aligned}\sigma^2 \operatorname{Tr}(H_J) &= \sigma^2 \operatorname{Tr}\big(W_J(W_J'W_J)^{-1}W_J'\big) \\ &= \sigma^2 \operatorname{Tr}\big(W_J'W_J(W_J'W_J)^{-1}\big) \quad (\text{using } \operatorname{Tr}(AB) = \operatorname{Tr}(BA)) \\ &= \sigma^2 \operatorname{Tr}(I_J) = \sigma^2 J\end{aligned}$$

Degrees of Freedom for Series Regression

$$\operatorname{df}(\hat{m}_J) = \operatorname{Tr}(H_J) = J$$

The degrees of freedom equal the number of basis functions J — the number of "effective parameters." This is why J controls model complexity.

The IMSE for series regression with J basis functions is therefore:

$$\operatorname{IMSE}(\hat{m}_J, m) = -N\sigma^2 + \sum_{i=1}^{N} \mathbb{E}[(Y_i - \hat{m}_J(x_i))^2] + 2\sigma^2 J$$

40 Mallows C_J Criterion and Empirical Risk Minimization

40.1 Constructing an Unbiased Risk Estimate

The IMSE formula requires knowledge of σ^2. Let $\hat{\sigma}^2$ be an unbiased estimate of σ^2 (discussed below).

Definition 40.1 (Mallows C_J Criterion)**.** For a series regression with J basis functions, the **Mallows C_J criterion** is:

$$C_J = -N\hat{\sigma}^2 + \underbrace{\sum_{i=1}^{N}(Y_i - \hat{m}_J(x_i))^2}_{\text{RSS}_J} + 2\hat{\sigma}^2 J$$

where $\text{RSS}_J = \sum_i (Y_i - \hat{m}_J(x_i))^2$ is the **Residual Sum of Squares**.

Taking expectations and using $\mathbb{E}[\hat{\sigma}^2] = \sigma^2$:

$$\mathbb{E}[C_J] = -N\sigma^2 + \sum_{i=1}^{N} \mathbb{E}[(Y_i - \hat{m}_J(x_i))^2] + 2\sigma^2 J = \operatorname{IMSE}(\hat{m}_J, m)$$

Mallows C_J is an Unbiased Risk Estimator

$$\mathbb{E}[C_J] = \text{IMSE}(\hat{m}_J, m)$$

C_J is an unbiased estimate of the IMSE/Risk for the J-basis-function series regression.

40.2 Empirical Risk Minimization: Choosing J^*

We want to select the number of basis functions J to minimize risk. Since IMSE is not directly observable (it depends on the unknown m), we minimize C_J instead:

Empirical Risk Minimization (Model Selection)

We consider models indexed by $J \in \{1, 2, \ldots, L\}$ where $L \ll N$ is the maximum model complexity we consider (e.g., $L = 1000$, $N = 10000$).

1. **Estimate $\hat{\sigma}^2$:** Fit the "big" model with L basis functions. Use:

$$\hat{\sigma}^2 = \frac{1}{N-L} \sum_{i=1}^{N} (Y_i - \hat{m}_L(x_i))^2$$

   The large model (L basis functions) has low bias so $\mathbb{E}[\hat{\sigma}^2] \approx \sigma^2$.

2. **Fit all candidate models:** For each $J = 1, 2, \ldots, L$, compute $\hat{m}_J$ (OLS of Y onto W_J) and the corresponding C_J.

3. **Choose J^*:**

$$J^* = \arg\min_{J \in \{1,\ldots,L\}} C_J$$

   Use $\hat{m}_{J^*}$ as the final estimator.

Intuition for the C_J vs. J plot:

- For small J: high RSS (large apparent error, high bias) $\Rightarrow$ large C_J.
- For large J: RSS decreases but the penalty $2\hat{\sigma}^2 J$ increases $\Rightarrow$ large C_J.
- The minimum of C_J occurs at an intermediate J^* that balances fit and complexity.
- The model with the lowest estimated risk is (ideally) the model with the lowest true risk.

Degrees of freedom as effective parameters: Intuitively, $\text{df}(\hat{m}) = \text{Tr}(H)$ counts the *effective number of parameters* being fit. For a J-term series regression, this equals exactly J. For more complex estimators (like kernel regression or trees), it is harder to compute analytically.

41 Kernel Regression

41.1 The Kernel Estimator

Kernel regression is a local averaging method that estimates $m(x)$ by taking a weighted average of Y_i values near x:

$$\hat{m}(x) = \sum_{i=1}^{N} W_i(x) Y_i$$

where the weights are:

$$W_i(x) = \frac{K\left(\dfrac{x - x_i}{h}\right)}{\sum_{j=1}^{N} K\left(\dfrac{x - x_j}{h}\right)}$$

and $K(\cdot)$ is a **kernel function** and $h > 0$ is the **bandwidth** (smoothing parameter).

A common kernel is the **Epanechnikov kernel**:

$$K(u) = \begin{cases} \frac{3}{4}(1 - u^2) & \text{if } |u| \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

which is zero for $|u| > 1$ and satisfies $\lim_{u \to \infty} K(u) = 0$. The kernel is similar to nearest-neighbors: observations far from x (beyond bandwidth h) get zero weight.

41.2 The Smoothing Matrix

Kernel regression is a linear estimator. Define the $N \times N$ **smoothing matrix**:

$$H_h = \begin{pmatrix} W_1(x_1) & W_2(x_1) & \cdots & W_N(x_1) \\ W_1(x_2) & W_2(x_2) & \cdots & W_N(x_2) \\ \vdots & & \ddots & \vdots \\ W_1(x_N) & W_2(x_N) & \cdots & W_N(x_N) \end{pmatrix}$$

so that $\hat{m} = H_h \cdot Y$ at the design points. The degrees of freedom are $\mathrm{df}(\hat{m}) = \mathrm{Tr}(H_h)$.

41.3 Effect of Bandwidth h

The bandwidth h is the model complexity parameter analogous to J in series regression:

- $h \to \infty$: All weights $W_i(x)$ become nearly equal ($\approx 1/N$). The estimator converges to the constant $\bar{Y}$ (zero df, high bias, low variance).
- $h \to 0$: Each $\hat{m}(x_i) \approx Y_i$ (correct-the-dots). Full N degrees of freedom, zero bias, infinite variance.
- Small h: Weighted local averages — picks up local structure but noisy.

The optimal bandwidth is chosen by minimizing $C_h = -N\hat{\sigma}^2 + \|Y - H_h Y\|^2 + 2\hat{\sigma}^2 \,\mathrm{Tr}(H_h)$.

42 Stein's Unbiased Risk Estimate (SURE)

42.1 The Problem: Non-Linear Estimators

For non-linear estimators (where $\hat{m}$ is a nonlinear function of Y, such as trees, LASSO, or ridge regression), we cannot use $\mathrm{Tr}(H)$ directly. Stein (1981) provides a general formula.

42.2 Stein's Lemma

The key tool is Stein's identity for Gaussian random variables:

Theorem 42.1 (Stein's Identity, 1981)**.** *Let $u \sim N(0, \sigma^2 I_N)$. For a function $g : \mathbb{R}^N \to \mathbb{R}^N$ satisfying mild regularity conditions:*

$$\mathbb{E}[g_i(Y)(Y_i - m(x_i))] = \sigma^2 \mathbb{E}\left[\frac{\partial g_i(Y)}{\partial Y_i}\right]$$

where g_i is the i-th component of g.

42.3 Deriving Degrees of Freedom via Stein

Define $g(Y) = \hat{m}(Y) - Y$ (estimator minus data). Then:

$$\begin{aligned}
\mathrm{Cov}(\hat{m}(x_i), Y_i) &= \mathbb{E}[\hat{m}(x_i)(Y_i - m(x_i))] \\
&= \mathbb{E}[(\hat{m}(x_i) - Y_i)(Y_i - m(x_i))] + \mathbb{E}[Y_i(Y_i - m(x_i))] \\
&= \mathbb{E}[g_i(Y)(Y_i - m(x_i))] + \sigma^2 \\
&= \sigma^2 \mathbb{E}\left[\frac{\partial g_i(Y)}{\partial Y_i}\right] + \sigma^2 \quad \text{(by Stein's identity)} \\
&= \sigma^2 \mathbb{E}\left[\frac{\partial \hat{m}(x_i)}{\partial Y_i} - 1\right] + \sigma^2 \\
&= \sigma^2 \mathbb{E}\left[\frac{\partial \hat{m}(x_i)}{\partial Y_i}\right]
\end{aligned}$$

Degrees of Freedom via Stein (SURE)

For any estimator $\hat{m}$ (not just linear), provided $u_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2)$:

$$\mathrm{df}(\hat{m}) = \sum_{i=1}^{N} \mathbb{E}\left[\frac{\partial \hat{m}(x_i)}{\partial Y_i}\right]$$

This is the expected sum of partial derivatives of the estimator with respect to its inputs — a measure of how sensitive the fitted values are to perturbations in the data.

Connection to the hat matrix: For a linear estimator $\hat{m} = HY$, we have $\partial\hat{m}(x_i)/\partial Y_i = h_{ii}$, so $\mathrm{df}(\hat{m}) = \sum_i h_{ii} = \mathrm{Tr}(H)$, consistent with our earlier result.

42.4 SURE for General Estimators

The full SURE formula is:

Definition 42.2 (Stein's Unbiased Risk Estimate (SURE), 1981)**.** For a tuning-parameter-indexed estimator $\hat{m}_\lambda$ (where λ controls complexity):

$$\mathrm{SURE}(\hat{m}_\lambda, m) = -N\sigma^2 + \sum_{i=1}^{N}(Y_i - \hat{m}_\lambda(x_i))^2 + 2\sigma^2 \sum_{i=1}^{N} \frac{\partial \hat{m}_\lambda(x_i)}{\partial Y_i}$$

$$\mathbb{E}[\mathrm{SURE}(\hat{m}_\lambda, m)] = \mathrm{IMSE}(\hat{m}_\lambda, m)$$

Jones–Stein estimator: A celebrated application is the James–Stein estimator, which is a shrinkage estimator of a multivariate normal mean. Stein showed that the MLE (OLS) is inadmissible in dimension ≥ 3: there exist estimators with uniformly lower IMSE.

42.5 Ye (1998) Algorithm for Estimating Degrees of Freedom

For complicated nonlinear estimators where $\partial\hat{m}(x_i)/\partial Y_i$ is hard to compute analytically, Ye (1998, JASA) provides a simulation-based algorithm:

Ye (1998) Algorithm for Estimating $\mathrm{df}(\hat{m}_\lambda)$

Tuning parameter λ (e.g., number of leaves in a tree). Choose hyperparameter h for perturbation size ($h = 0.6\hat{\sigma}$ recommended).

1. **For** $s = 1, \ldots, S$:
   (a) Draw perturbation: $D_i^{(s)} \overset{\text{i.i.d.}}{\sim} N(0, h^2)$, $i = 1, \ldots, N$.
   (b) Fit $\hat{m}_\lambda^{(s)}$ using perturbed data $Y_i + D_i^{(s)}$.
2. **For** $i = 1, \ldots, N$:
   (a) Regress $\hat{m}_\lambda^{(s)}(x_i)$ onto $(1, D_i^{(s)})$ using $s = 1, \ldots, S$ simulation draws.
   (b) Let $\hat{\delta}_i$ be the coefficient on $D_i^{(s)}$.
3. **Construct:**
$$\widehat{\mathrm{df}}(\hat{m}_\lambda) = \sum_{i=1}^{N} \hat{\delta}_i$$
   because $\hat{\delta}_i \approx \partial\hat{m}_\lambda(x_i)/\partial Y_i$ by the finite-difference approximation.

The risk estimate is then $\mathrm{SURE}(\hat{m}_\lambda, m) = -N\hat{\sigma}^2 + \mathrm{RSS} + 2\hat{\sigma}^2\,\widehat{\mathrm{df}}(\hat{m}_\lambda)$.

43 Regression Trees

43.1 Motivation

Regression trees (Leo Breiman) provide a flexible, nonlinear, nonparametric estimator that captures non-linear and clustered trends in data — similar to k-means and kNN in spirit.

43.2 Structure

Definition 43.1 (Regression Tree)**.** A regression tree recursively partitions the feature space $\mathcal{X}$ into rectangular "cells" (leaves). Given $x_i = (W_i, Z_i)'$:

- $W_i \in \mathcal{W} = [\underline{w}, \bar{w}]$: one feature.
- $Z_i \in \mathcal{Z} = [\underline{z}, \bar{z}]$: another feature.

The tree splits the feature space recursively. For example, with thresholds w_1 and z_1, z_2:

- Root splits on W_i: left ($W_i \leq w_1$) vs. right ($W_i > w_1$).
- Each subtree splits further on Z_i: creating leaf nodes $\mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3, \mathcal{X}_4$.

Cell mean estimator: If $x_i \in \mathcal{X}_j$ (leaf j), the prediction is the sample mean of Y values in that leaf:

$$\hat{m}(x_i) = \frac{\sum_{k=1}^{N} Y_k \cdot \mathbb{1}(x_k \in \mathcal{X}_j)}{\sum_{k=1}^{N} \mathbb{1}(x_k \in \mathcal{X}_j)} \qquad (\text{"cell mean"})$$

At each split, we choose the threshold that minimizes the within-leaf squared residuals.

43.3 Overfitting and Regularization

Trees are prone to **overfitting**: if we keep partitioning, we eventually get one leaf per observation, achieving zero in-sample error (correct-the-dots). This gives an RSS $\to 0$ but df $\to N$ and IMSE $\to$ large.

The risk as a function of number of leaves J follows a U-shape: decreasing initially (lower bias) then increasing (higher variance from overfitting). The optimal J^* minimizes C_J (or SURE).

Heuristics to prevent overfitting:

(i) Choose a fixed maximum number of leaves J.

(ii) Require each leaf to contain a minimum number of observations.

The number of parameters is much more complex than the number of leaves (each split involves choosing a variable and a threshold). Using each data point as its own node gives very bad out-of-sample fit.

The C_J vs. J plot for trees: Plotting C_J against the number of leaves J:

- J small: High RSS (underfitting, large bias) $\Rightarrow$ large C_J.
- J optimal: C_J is minimized. Choose this J^*.
- J large: Low RSS but high df penalty $\Rightarrow$ large C_J (overfitting).

44 Summary Table

Object	Formula	Interpretation
Apparent error	$\mathbb{E}[\|Y - \hat{m}\|^2]$	Expected in-sample fit
Noise floor	$N\sigma^2$	Irreducible error
IMSE	$\sum_i \mathbb{E}[(\hat{m}(x_i) - m(x_i))^2]$	Overall quality of estimator
Master formula	IMSE $= -N\sigma^2 +$ Apparent Error $+ 2\sigma^2 \operatorname{df}(\hat{m})$	Risk decomposition
df (linear)	$\operatorname{df}(\hat{m}) = \operatorname{Tr}(H)$	Sum of diagonal of hat matrix
df (series)	$\operatorname{df}(\hat{m}_J) = J$	Equals number of basis functions
df (Stein)	$\operatorname{df}(\hat{m}) = \sum_i \mathbb{E}[\partial \hat{m}(x_i)/\partial Y_i]$	General formula via differentiation
Mallows C_J	$C_J = -N\hat{\sigma}^2 + \text{RSS}_J + 2\hat{\sigma}^2 J$	Unbiased IMSE estimator
SURE	$-N\sigma^2 + \text{RSS} + 2\sigma^2 \sum_i \partial \hat{m}(x_i)/\partial Y_i$	Unbiased risk for any estimator
Kernel	$\hat{m}(x) = \sum_i W_i(x) Y_i,\ W_i \propto K((x - x_i)/h)$	Local averaging estimator
Tree cell mean	$\hat{m}(x_i) = \bar{Y}_{\mathcal{X}_j}$	Average over leaf containing x_i

Big Picture

The central problem of model selection is choosing complexity to balance in-sample fit (low apparent error) against model complexity (degrees of freedom). The master formula IMSE $= -N\sigma^2 +$ Apparent Error $+ 2\sigma^2 \operatorname{df}(\hat{m})$ quantifies this tradeoff ex-

actly. Mallows C_J provides an unbiased estimate of IMSE for series regression (where $\mathrm{df} = J$). Stein's SURE generalizes this to arbitrary estimators using the gradient formula $\mathrm{df}(\hat{m}) = \sum_i \mathbb{E}[\partial \hat{m}/\partial Y_i]$, with Ye's algorithm enabling simulation-based estimation when the gradient is intractable. Kernel regression and regression trees are two important nonlinear estimators whose complexity is controlled by bandwidth h and number of leaves J respectively — in both cases, C_J (or SURE) provides the right criterion for choosing the tuning parameter.